Division of Computing Science and Mathematics
Faculty of Natural Sciences
University of Stirling

Clinical NLP for Morbidity Extraction and NHS ICD-10 Coding:

Comparing Transformer-Centred Bio_ClinicalBERT and Terminology-Centred Apache cTAKES Pipelines on MIMIC-IV Discharge Summaries

Kristina Palesheva

Dissertation submitted in partial fulfilment for the degree of Master of Science in Big Data

September 2026

Abstract

Problem: Electronic health records contain rich information about patients' diseases and medical histories, but much of it remains buried in unstructured clinical text. Automatically converting this information into structured morbidity data and linking it to standardised NHS ICD-10 codes could make patient histories easier to retrieve, interpret and reuse.

***This study compares two contrasting approaches to this task**: a transformer-centred Bio_ClinicalBERT pipeline within EHRKit and a terminology-centred Apache cTAKES pipeline.*

Methodology: Bio_ClinicalBERT was developed using 100 manually annotated, patient-disjoint MIMIC-IV discharge summaries, while a further 60 records supported error-driven refinement of both pipelines. Once development was complete, both systems were frozen. A preliminary paired evaluation was conducted on 20 previously unseen records. Without any further modification, the frozen pipelines were subsequently evaluated on a separate set of 200 additional previously unseen discharge summaries, containing **2,103 gold-standard morbidity concepts**. The two evaluation sets were scored separately and were not pooled. Extracted morbidities were normalised using UMLS and SNOMED CT UK resources and linked to NHS ICD-10 5th Edition families.

Results: Although the absolute performance values changed in the larger evaluation, the direction of the precision–recall trade-off remained consistent. Bio_ClinicalBERT again achieved higher precision, whereas cTAKES again achieved higher recall. In the separate 200-record evaluation, Bio_ClinicalBERT achieved **90.07%** precision, **65.53%** recall and **75.86%** F1, while cTAKES achieved **71.62%** precision, **75.61%** recall and **73.56%** F1.

For ICD-10 assignment, cTAKES also achieved higher coding-suggestion accuracy (**92.47%** vs **88.91%**) and correct-code coverage (**87.41%** vs **83.28%**). Despite their different error profiles, **87.59%** of all gold-standard morbidities were detected by at least one of the two systems.

Conclusion: The larger evaluation reinforced the distinction between the two approaches: Bio_ClinicalBERT was consistently more selective and precise, whereas cTAKES captured a broader range of morbidities and performed better in terminology-based coding. *Their complementary behaviour suggests that combining transformer-based contextual recognition with terminology-driven extraction and normalisation may provide a more complete and reliable solution than either pipeline alone.*

Attestation

I understand the nature of plagiarism, and I am aware of the University's academic integrity policy.

I certify that this dissertation reports work completed by me during the MSc dissertation, except where external data, software, pre-trained models and terminology resources are explicitly acknowledged and cited.

The project used the MIMIC-IV-Note dataset [1], Apache cTAKES [5], Bio_ClinicalBERT [6], EHR-Kit [7], medSpaCy and ConText [11], [17], UMLS [4], SNOMED CT UK and NHS ICD-10 mapping resources [2], [3], [16]. These resources were not created by me and were used under their applicable access conditions and licences.

My contribution included the selection and preparation of the clinical records, manual morbidity annotation, development and refinement of the Bio_ClinicalBERT-centred pipeline, configuration and extension of the cTAKES-centred pipeline, terminology and ICD-10 mapping integration, construction and review of the evaluation data, implementation of the evaluation procedure, and analysis and interpretation of the results.

OpenAI Codex was used to assist with selected programming, debugging and explanation of technical issues. All AI-assisted outputs were reviewed, tested and adapted by me before inclusion. AI-assisted code and the corresponding prompts are identified and submitted in accordance with the MSc dissertation AI policy.

Signature **Date 27.08.2026**

Acknowledgements

I would like to express my sincere gratitude to Professor Bouamrane for their invaluable guidance, encouragement and support throughout this project. I am especially thankful for their thoughtful, highly professional and consistently relevant advice, which helped me to refine both the direction of the research and the way in which the work was presented. Their support throughout the project was extremely important to me.

I am particularly grateful for their suggestion to extend the work by linking extracted morbidities to standardised ICD-10 codes, which became an important and rewarding part of the dissertation.

I also deeply appreciate the confidence they placed in me by entrusting me with such a meaningful and challenging project. This opportunity allowed me to gain valuable knowledge, confidence and experience, for which I am sincerely thankful.

Consent for the University to Share your Dissertation

Please sign your name underneath one of the options below, to indicate whether you consent to the University sharing your dissertation (e.g. as an example to future students):

The University may make this dissertation, with indicative grade, available to others.

Kristina Palesheva

The University may make this dissertation available to others, but the grade may not be disclosed.

The University may not make this dissertation available to others.

Table of Contents

List of Figures

1 Introduction

1.1 Background and Context

Electronic health records (EHRs) contain both structured data, such as laboratory results and medication orders, and unstructured clinical narratives written by healthcare professionals. These narratives provide detailed information about patients' diagnoses, previous medical conditions, treatments and clinical progress. MIMIC-IV discharge summaries [1] are particularly valuable because they consolidate information from an entire hospital admission. However, relevant disease information may be distributed across several sections, expressed using different terminology or repeated multiple times. Consequently, reconstructing a patient's morbidity history through manual review can be time-consuming.

Natural language processing (NLP) offers transformation free-text clinical records into structured data that can be retrieved, compared and analysed more efficiently. In this project, morbidity extraction refers to the identification of diseases and pathological conditions documented as affecting the patient. This task involves more than detecting words that resemble disease names. Clinical narratives frequently contain abbreviations, spelling variations, misspellings, incomplete sentences and non-standard expressions. For example, *HTN, hypertension and high blood pressure may all describe the same clinical concept*.

Clinical context introduces an additional challenge because the occurrence of a disease term does not necessarily indicate that the patient has that disease. A mention may be negated, uncertain, hypothetical or associated with a family member. For example, *pneumonia* should be retained in "the patient was treated for pneumonia" but excluded in "*no evidence of pneumonia"*. Similarly, "*the patient's mother has diabetes*" describes family history rather than a morbidity of the patient. A reliable extraction system must therefore combine morbidity detection with assertion and contextual processing.

Extracted disease descriptions also need to be standardised. SNOMED CT provides a detailed clinical terminology for representing information in electronic records [2], whereas ICD-10 is primarily a classification used to categorise diseases for reporting, statistical analysis and other secondary purposes [3]. NHS Digital distinguishes SNOMED CT as a structured clinical vocabulary from ICD-10 as a statistical classification of diagnostic information [2], [3]. Linking different clinical expressions to a common concept and subsequently to a suggested NHS ICD-10 5th Edition category can reduce ambiguity and facilitate consistent reuse of the extracted information. However, automatically generated codes should be treated as suggestions rather than replacements for formal clinical coding.

Clinical information extraction can be implemented using terminology-driven systems or contextual language models. Apache cTAKES is an open-source natural language processing system for the clinical domain that generates linguistic and semantic annotations and supports the recognition of clinical entities and their mapping to UMLS concepts [4], [5]. Its terminology-based architecture may offer broad concept coverage and interpretable code suggestions, although its performance depends on dictionary coverage, pipeline configuration and assertion processing.

In contrast, models based on the Transformer architecture employ a different approach, learning to generate contextual language representations. The Bio_ClinicalBERT model was pre-trained on clinical texts and it can be fine-tuned for tasks such as named entity recognition in clinical notes [6]. Such models may be better able to interpret a term from the surrounding context, but they require task-specific annotated data and do not automatically provide standard medical codes. EHRKit provides a Python-based environment for applying different NLP functions and neural approaches to clinical text [7]. Terminology-driven and transformer-based approaches have different potential strengths, and their relative suitability for document-level morbidity extraction requires direct evaluation.

This project compares a Bio_ClinicalBERT-centred pipeline implemented within EHRKit with a terminology-centred cTAKES pipeline. Both systems process the same complete MIMIC-IV discharge summaries and produce concept-level morbidity lists with suggested NHS ICD-10 5th Edition families.

The purpose of the project is to compare the extraction and coding performance of the two approaches, identify their complementary strengths and error patterns, and assess the potential value of a hybrid morbidity-extraction system.

1.2 Research questions

The project addresses the following research questions:

1) How accurately can the fine-tuned Bio_ClinicalBERT pipeline implemented within EHRKit and an Apache cTAKES-centred pipeline extract unique morbidity concepts from complete MIMIC-IV discharge summaries in the MIMIC-IV dataset?
2) How effectively can the morbidities extracted by each pipeline be assigned appropriate suggested NHS ICD-10 5th Edition families?
3) What complementary strengths, limitations and error patterns do the transformer-centred and terminology-centred approaches demonstrate, and what are the implications for the design of a hybrid morbidity-extraction system?

1.3 Scope and Objectives

The project focuses on extracting diseases and pathological conditions from complete MIMIC-IV discharge summaries. The unit of evaluation is a unique morbidity concept within an individual document. Repeated mentions, abbreviations and synonymous descriptions representing the same condition are consolidated and counted only once. Confirmed current and historical patient conditions are included, while negated, uncertain, hypothetical, conditional, family-related and non-diagnostic mentions are excluded.

The scope does not include the extraction of medications, dosages, clinical procedures or laboratory results. ICD-10 outputs are evaluated as automated suggestions and are not intended to replace coding performed by qualified clinical coders.

The study uses patient-disjoint data across three experimental phases. A corpus of 100 manually annotated discharge summaries supports Bio_ClinicalBERT development, 60 additional records support iterative refinement of both pipelines, and, after both configurations were frozen, 20 previously unseen records support a preliminary paired evaluation followed by an expanded evaluation on 200 additional previously unseen records.

The objectives of the project were:

- To prepare complete MIMIC-IV discharge summaries and create occurrence-level development annotations and concept-level final gold standards for morbidity extractions.
- To develop, fine-tune and refine a Bio_ClinicalBERT morbidity-extraction pipeline within the EHRKit environment.
- To configure and refine an Apache cTAKES-centred pipeline for extracting morbidities from the same type of discharge summaries.

- To apply context filtering and deduplication within a record so that both systems produce comparable lists of unique patient morbidities.
- To normalise extracted morbidities using UMLS and SNOMED CT UK resources and assign suggested NHS ICD-10 5th Edition families.
- To compare the two frozen pipelines on the same preliminary and expanded evaluation records using precision, recall and F1 score.
- To evaluate ICD-10 coding coverage, correct-code coverage and coding-suggestion accuracy, and to analyse the principal strengths and limitations of each approach.

1.4 Main contribution

The main contribution of this dissertation is a **controlled comparison of transformer-centred and terminology-centred approaches to document-level morbidity extraction**. Both complete pipelines were evaluated on the same held-out MIMIC-IV discharge summaries using a common concept-level gold standard, duplicate-exclusion policy and scoring procedure.

The project also produced a patient-disjoint annotated corpus and an auditable end-to-end workflow covering morbidity extraction, contextual filtering, concept-level deduplication, terminology normalisation and suggested NHS ICD-10 5th Edition coding. This extends the comparison beyond entity recognition by examining whether extracted morbidity information can be converted into a standardised and reusable representation.

The comparison revealed complementary and consistent system behaviour across the expanded evaluation. *The Bio_ClinicalBERT-centred pipeline achieved substantially higher precision, while the cTAKES-centred pipeline achieved higher recall and stronger ICD-10 coding performance. Although Bio_ClinicalBERT achieved a modestly higher F1 score in the expanded 200-record evaluation, neither approach was uniformly superior across the evaluated dimensions. The persistence of these contrasting error profiles at larger scale further motivates future evaluation of a hybrid pipeline combining terminology-based candidate coverage with transformer-based contextual validation*.

1.5 Overview of dissertation

Chapter 2 reviews related work on clinical concept extraction, contextual processing, transformer models, terminology normalisation and automatic medical coding, and identifies the research gap addressed by the dissertation.

Chapter 3 presents the experimental methodology, including data preparation, annotation, partitioning, refinement and evaluation design.

Chapter 4 describes the implementation of the Bio_ClinicalBERT-centred and cTAKES-centred pipelines.

Chapter 5 reports the development-stage, preliminary and expanded evaluation results.

Chapter 6 evaluates the project against its research questions and objectives, compares the findings with existing work, discusses limitations and proposes future development.

2 State-of-the-Art

Research on extracting information about morbidity from clinical texts covers several related but distinct tasks:

- Detecting disease mentions;
- Determining their clinical context;
- Normalising alternative expressions to standard concepts;
- Assigning classification codes.

Previous studies often evaluate these tasks independently. However, producing a structured morbidity summary requires them to operate as a connected pipeline. This chapter reviews the approaches most relevant to that process and identifies the gap addressed by this dissertation.

2.1 Terminology-driven clinical concept extraction

Early clinical information extraction systems relied primarily on dictionaries, medical terminologies and manually defined linguistic rules. These approaches identify text matching known clinical expressions and associate the extracted mention with a concept from a terminology such as the Unified Medical Language System (UMLS). Their main advantage is that entity recognition and concept normalisation can be performed together, producing interpretable concept identifiers without requiring a large task-specific training dataset.

Apache cTAKES is one of the most mature open-source systems in this category. It was developed as a modular natural language processing (NLP) pipeline for the clinical domain, based on the Unstructured Information Management Architecture (UIMA) and includes components for sentence segmentation, tokenisation, part-of-speech tagging, clinical entity recognition, UMLS concept mapping and assertion processing [5]. These contextual attributes allow extracted concepts to be classified as current, historical, possible, related to family history, or negated.

The terminology-driven design of cTAKES offers several advantages for morbidity extraction. It can recognise concepts from a large clinical vocabulary, provide UMLS identifiers and generate outputs that can be inspected at each processing stage. It can also be applied without first training an entity-recognition model on a large local corpus. However, its accuracy depends on the coverage of its dictionaries, the quality of lexical matching and the configuration of its processing components. Ambiguous abbreviations, misspellings and non-standard expressions may be incorrectly mapped or missed, while broad dictionary matching can generate concepts that are present in the text but do not represent confirmed patient morbidities.

Lossio-Ventura et al. compared cTAKES with CLAMP, MetaMap, NCBO Annotator, QuickUMLS and ScispaCy using i2b2 and MIMIC-III records [8]. Their results showed substantial variations across systems and evaluation conditions. For example, performance differed between exact and inexact matching and across challenges such as negation, abbreviations, ambiguity and misspelling. No system was consistently optimal for all extraction problems. This demonstrates that the suitability of a clinical NLP system depends on the specific entity definition, dataset and evaluation method.

Other systems combine terminology resources with statistical learning. MedCAT, for example, uses self-supervised learning to recognise clinical concepts and map them to standard terminologies such as UMLS or SNOMED CT. It has been evaluated on large collections of clinical records from UK hospitals [9]. This work demonstrates the potential benefit of combining contextual representations with medical terminologies. However, effective local adaptation of MedCAT may require a substantial collection of domain-specific clinical text, and *its reported* *evaluations do not directly answer how cTAKES compares with a task-specific Bio_ClinicalBERT pipeline for unique morbidity extraction from complete MIMIC-IV discharge summaries.*

2.2 Assertion and contextual processing

Clinical entity recognition alone is insufficient for constructing a morbidity history. A document may mention a condition that is denied, is hypothetical or uncertain in nature, relates to the past, or concerns another person rather than the patient. Systems that do not process these properties may achieve high mention coverage while generating clinically misleading false positives.

NegEx introduces a rule-based method in which lexical triggers such as ‘no’, ‘denies’ and ‘ruled out’ determine whether a nearby clinical concept is negated [10]. ConText extends this approach to additional contextual properties, including temporality, hypothetical status and experiencer [11]. Harkema et al. found that surface-level contextual rules could achieve useful performance across several report types. However, they also concluded that some distinctions, particularly between recent and historical conditions, require information beyond local lexical triggers.

Previous evaluation studies have generally measured assertion processing at the individual mention level. **The present project instead requires contextual decisions to contribute to a document-level list of unique patient morbidities**. This introduces an additional aggregation problem because different mentions of the same disease may have different contextual properties across a document. For example, a condition appearing in a Family History or Discharge Instructions section does not necessarily represent a confirmed morbidity of the patient.

2.3 Transformer-based clinical concept extraction

The introduction of transformer models changed clinical concept extraction by enabling systems to learn context-dependent representations instead of relying exclusively on dictionary matching. **BioBERT** demonstrated that continuing the pre-training of the BERT model on biomedical publications improves performance on named entity recognition, relation extraction, and question-answering tasks in the biomedical domain. [12]. However, biomedical publications differ from clinical records in their vocabulary, structure and writing style.

Bio_ClinicalBERT addressed this domain difference by adapting a biomedical BERT model to clinical notes. Alsentzer et al. released models trained on general clinical text and discharge summaries and found that clinical-domain pretraining improved performance on several, although not all, evaluated clinical NLP tasks [6]. These findings indicate that domain-specific pretraining is beneficial but does not remove the need for task-specific fine-tuning and validation.

Nevertheless, benchmark performance does not transfer automatically to document-level morbidity extraction. Public clinical NLP datasets use different entity definitions, document types and annotation policies. Many evaluate individual spans rather than unique concepts aggregated across a complete record. Transformer models also produce entity spans rather than standard terminology identifiers, meaning that concept normalisation must be performed separately.

Complete discharge summaries present an additional technical challenge because their length commonly exceeds the input limit of BERT-based models. The text must therefore be divided into sections or overlapping chunks. Pascual et al. similarly identified long-document processing as a principal limitation of BERT-based automatic ICD coding [13]. Chunking can separate a disease mention from relevant contextual information and can create duplicate predictions in overlapping text segments.

EHRKit provides a Python environment that integrates multiple NLP libraries and supports a range of EHR-processing tasks [7]. It should be distinguished from an extraction model: EHRKit provides the environment in which a model can be implemented, whereas extraction performance is determined by the selected model, its training data and the surrounding processing rules. Consequently, **the comparison in this dissertation is specifically between a fine-tuned Bio_ClinicalBERT pipeline implemented within EHRKit and a configured cTAKES pipeline**, rather than between EHRKit and cTAKES as equivalent algorithms.

2.4 Concept normalisation and medical coding

Clinical concept normalisation maps different textual expressions to a common medical concept. *Terminology-driven systems such as cTAKES perform this operation by associating matched expressions with UMLS Concept Unique Identifiers. Transformer-based entity-recognition models do not normally provide this link directly; thus, a separate normalisation stage is required.*

Concept normalisation should be distinguished from automatic ICD coding. In most studies on automated coding, the assignment of ICD codes is treated as a multi-label document classification task. For example, Mullenbach et al. developed the CAML architecture, which uses label-specific attention to predict ICD codes from complete clinical notes and highlight relevant text for each prediction [14]. This approach predicts the collection of codes associated with the document but does not necessarily extract and validate each individual morbidity mention before coding it.

Transformer-based coding methods face difficulties associated with long documents, large label spaces and rare codes, meaning that they must select from a large number of possible ICD codes, while many uncommon codes are represented by only a small number of examples in the training data. Pascual et al. found that pre-trained transformers could achieve competitive results using selected portions of clinical documents but that aggregating information across long records remained a major limitation [13]. *This highlights the continuing need for controlled evaluation using manually reviewed clinical data.*

Mapping from SNOMED CT to ICD-10 is also not a simple one-to-one lookup. The two systems serve different purposes and represent information at different levels of detail. NHS guidance states that national cross-maps are only semi-automated because additional information from the EHR may be required before a final classification code can be assigned [2]. SNOMED International similarly uses complex and extended map reference sets when contextual rules are required to determine an ICD-10 target [16].

Therefore, coding evaluation should assess:

- **ICD-10 coding coverage** — the proportion of correctly extracted morbidities that receive an ICD-10 suggestion;
- **coding-suggestion accuracy** — the proportion of suggestions assigned to correctly extracted morbidities that are correct; and
- **correct-code coverage** — the proportion of correctly extracted morbidities that receive the correct ICD-10 suggestion.

These distinctions motivate the use of coding coverage, correct-code coverage and coding-suggestion accuracy in addition to precision, recall and F1.

2.5 Comparison of related work

Table 2.1 Comparison of clinical NLP approaches

Study or system	Approach	Main evaluation focus	Limitation relative to this project
Savova et al. [5]	Modular terminology-driven cTAKES pipeline	Clinical entity recognition and linguistic annotation	Does not provide a direct comparison with a fine-tuned clinical transformer on MIMIC-IV
Harkema et al. [11]	Rule-based contextual processing	Negation, temporality, hypothetical status and experiencer	Evaluates contextual properties separately from end-to-end morbidity normalisation and coding
Lossio-Ventura et al. [8]	External comparison of six concept-recognition systems	Multiple entity types and extraction challenges on i2b2 and MIMIC-III	Uses mention-level evaluation and does not compare cTAKES with a task-specific Bio_Clinical-BERT model
Kraljevic et al. [9]	Self-supervised concept recognition with UMLS/SNOMED CT	Clinical concept extraction across large hospital datasets	Requires substantial local training data and does not evaluate the specific cTAKES - ClinicalBERT comparison
Alsentzer et al. [6]	Clinical-domain BERT pretraining	General clinical NLP benchmark tasks	Does not evaluate unique morbidity aggregation or NHS ICD-10 mapping
Mullenbach et al. [14] and Pascual et al. [13]	Document-level automatic ICD coding	Multi-label prediction of codes from clinical notes	Predict codes directly rather than extracting, validating and deduplicating individual morbidities
Falis et al. [15]	Generative-model coding and data augmentation	ICD-10 coding of synthetic and real MIMIC-IV summaries	Limited reliability on real records and no task-specific morbidity gold standard

These studies are not directly comparable through their reported F1 scores because they use different corpora, annotation schemes, entity categories and matching criteria. Table 2.1 should therefore be used to compare research design and task coverage rather than to rank the systems quantitatively.

2.6 Identified Research Gap

Existing research demonstrates that terminology-driven systems can provide broad concept coverage and direct links to standard vocabularies, while transformer models can learn contextual patterns that are difficult to express through dictionaries and fixed rules. However, previous evaluations generally focus on one component of the problem: mention-level entity recognition, assertion classification, concept normalisation or document-level ICD prediction.

The reviewed literature does not establish how a fine-tuned Bio_ClinicalBERT morbidity-extraction pipeline compares with cTAKES when both systems are:

- applied to the same complete MIMIC-IV discharge summaries;
- evaluated using the same manually created gold standard;
- assessed at the level of unique morbidity concepts within each document;
- required to exclude mentions according to clinical context;
- evaluated for both morbidity extraction and suggested NHS ICD-10 coding.

This dissertation addresses that gap through a controlled comparison of the two approaches on the same previously unseen records. The evaluation separates extraction performance from coding performance and measures precision, recall, F1 score, ICD-10 coding coverage, correct-code coverage and coding-suggestion accuracy. It therefore examines not only whether the systems can detect disease expressions, but whether they can produce a reliable and standardised summary of a patient's morbidities.

The comparison also examines whether the two approaches produce complementary errors. This is important because similar aggregate performance can conceal different operating behaviour: one system may prioritise precision, while another may provide greater recall or terminology coverage. Identifying such trade-offs provides a basis for assessing whether a hybrid approach could combine the strengths of both systems.

3 Experimental Methodology

This chapter presents the methodology and experimental controls used to support a reproducible and fair comparison of the two morbidity-extraction pipelines. Their technical implementation and performance results are presented in Chapters 4 and 5, respectively.

3.1 Research design

This study uses a **controlled, paired comparison of two complete NLP pipelines for converting free-text discharge summaries into structured lists of patient morbidities**.

The first was a transformer-centred pipeline based on a fine-tuned **Bio_ClinicalBERT** sequence tagger [6] implemented within the EHRKit environment [7].

The second was a terminology-centred pipeline based on **Apache cTAKES** [5] and its UMLS-linked DiseaseDisorderMention annotations [4], [5]. The study evaluates the complete implemented workflows rather than the unmodified Bio_ClinicalBERT and cTAKES systems in isolation.

Development was iterative: supervised Bio_ClinicalBERT training and model selection were followed by error-driven refinement of the contextual, terminology, deduplication and mapping components of both workflows. *The refinement procedure is described in Section 3.5*.

The experiment consisted of three sequential phases. In **the first phase**, a patient-disjoint corpus of 100 manually annotated MIMIC-IV discharge summaries was used for Bio_ClinicalBERT development [1]. It was divided into 70 training notes, 10 calibration notes and 20 initial internal-validation notes. The training partition was used to optimise the model weights, while the calibration partition supported checkpoint and confidence-threshold selection. The initial internal-validation partition was used to assess the first complete version of the transformer-centred pipeline. cTAKES did not require supervised training on these annotations because its primary extraction process was based on terminology resources and assertion-processing components.

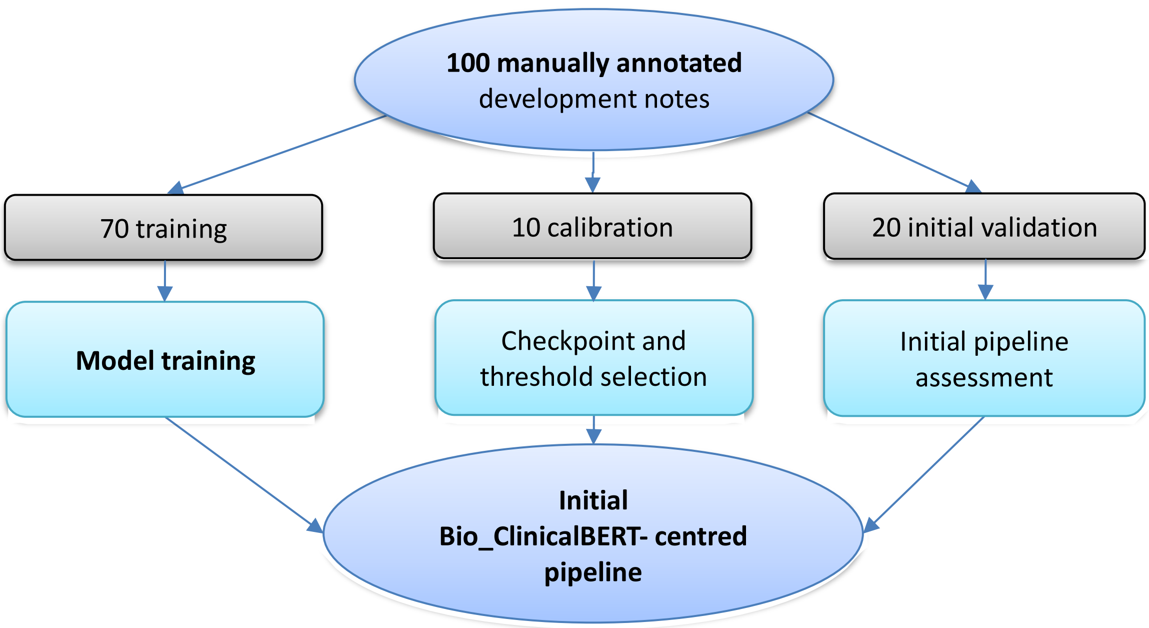

*Figure 3.1 **PHASE 1**: BIO_CLINICALBERT development*

The second phase used 60 additional discharge summaries from patients not represented in the first-phase partitions. The records supported manual error analysis and iterative refinement of both complete pipelines. Because their outputs informed subsequent development decisions, these records were treated as development data and were excluded from both paired evaluations.

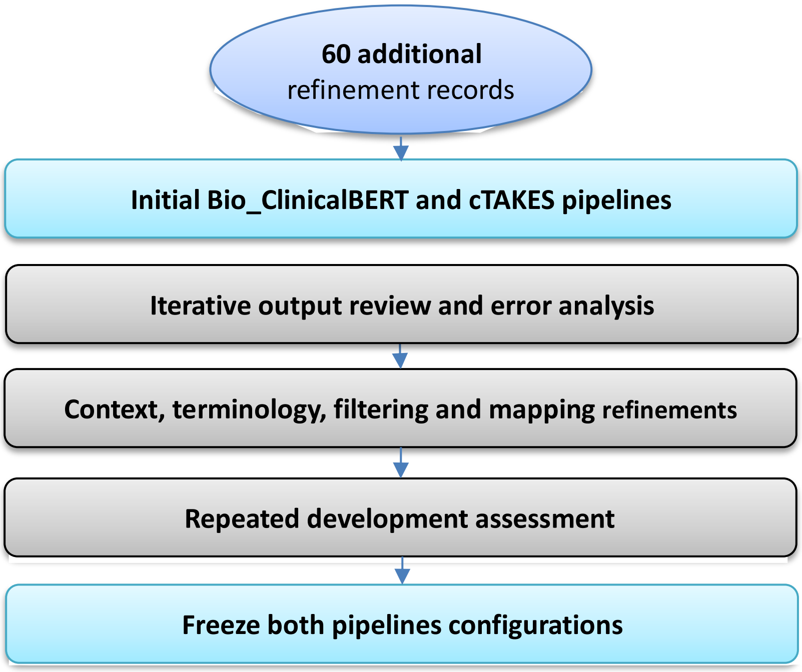

Figure 3.2 **PHASE 2**: Pipeline refinement

In **the third phase**, both pipeline configurations were frozen before any evaluation records were processed. A preliminary paired evaluation was first conducted on 20 previously unseen, patient-disjoint discharge summaries. This was subsequently followed by an expanded evaluation on 200 additional previously unseen, patient-disjoint discharge summaries. In particular, the preliminary 20-record results did not trigger any further changes to either pipeline before the expanded evaluation.

Both pipelines processed the same clinical texts within each evaluation and were compared against the same manually reviewed gold-standard morbidity concepts using identical concept-matching, duplicate-exclusion and scoring rules. The preliminary evaluation contained 274 unique gold-standard morbidity concepts, while the expanded evaluation contained 2,103.

The unit of evaluation remained a unique morbidity concept within a note rather than each individual textual occurrence of a disease. Repeated mentions and synonymous expressions representing the same condition were consolidated before scoring. This prevented frequently repeated conditions from having a disproportionate effect on the results.

Figure 3.1, *Figure 3.2*, and Figure 3.3 illustrate the separation between supervised Bio_ClinicalBERT development, iterative refinement of both end-to-end pipelines and evaluation of the frozen configurations. Patient-level separation across development, refinement and evaluation data reduced the risk of information leakage.

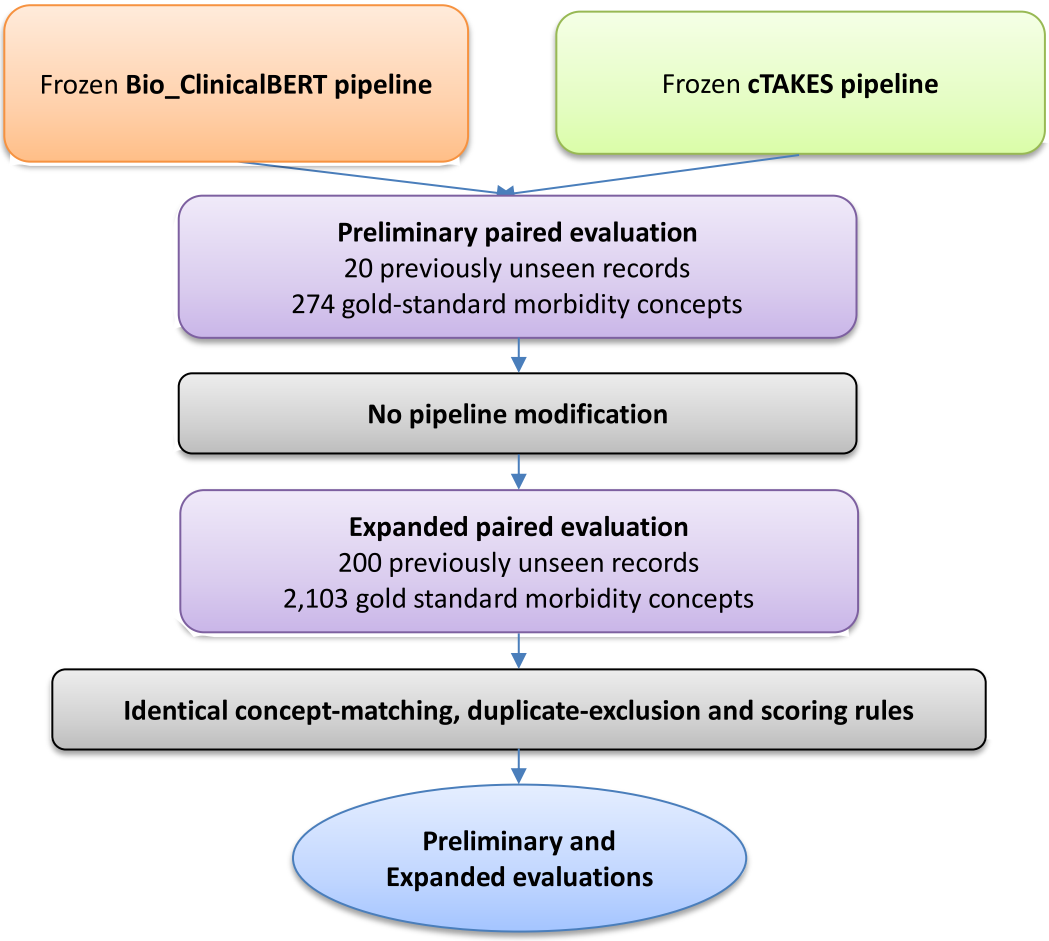

Figure 3.3 ***PHASE 3****: Preliminary and expanded paired evaluation of the frozen pipelines*

3.2 MIMIC-IV Dataset and record selection

The clinical text used in this project was obtained from MIMIC-IV-Note, a de-identified collection of free-text clinical documentation linked to the MIMIC-IV database [1]. Complete *discharge summaries were selected because they provide a consolidated account of a hospital admission, including diagnoses, previous medical history, investigations, treatment and discharge planning*. However, their length, repeated information, abbreviations and variation in section structure make them a challenging source for automated morbidity extraction. MIMIC-IV-Note contains de-identified clinical data. Access was obtained through PhysioNet under its credentialing and data-use requirements, and only the de-identified database identifiers were used during processing [1].

The study used 380 discharge summaries across the three experimental phases described in Section 3.1. The first 100 notes formed the Bio_ClinicalBERT development corpus, 60 additional notes were used for iterative refinement of both pipelines, 20 previously unseen notes formed the preliminary paired evaluation set, and a further 200 previously unseen notes formed the expanded evaluation set. Each selected note was associated with a note identifier, subject identifier and hospital-admission identifier.

Patient-level separation was applied across all development, refinement and evaluation partitions. Where multiple eligible records were available for one patient, only one discharge summary was retained. The preliminary and expanded evaluation records were selected only after excluding note, patient and admission identifiers represented in the development and refinement data, and the expanded evaluation additionally excluded patients and records represented in the preliminary evaluation.

The complete text of each selected discharge summary was retained, including its clinical sections and line structure. **During both paired evaluations, the two pipelines received the same discharge-summary text and record identifiers within each evaluation set**. CSV files were used as the cTAKES input, while Excel workbooks were used for structured review and evaluation. These formatting differences did not alter the clinical information available to either pipeline.

3.3 Manual annotation and gold-standard construction

3.3.1 Bio_ClinicalBERT development annotations

The Bio_ClinicalBERT development corpus contained 100 manually annotated discharge summaries. The notes were uploaded to Label Studio, where words and phrases referring to diseases or pathological conditions were highlighted and assigned the MORBIDITY label. After annotation, the labelled text spans, character positions and record identifiers were exported from Label Studio for data preparation and supervised model development.

The occurrence-level annotations were subsequently converted into BIO sequence labels. The beginning of a morbidity expression was represented as *B-MORBIDITY*, subsequent tokens within the same expression as *I-MORBIDITY*, and all other tokens as *O*. This representation allowed multi-word expressions such as *chronic kidney disease* to be learned as a single morbidity span.

The occurrence-level Label Studio annotations were used for model development and should be distinguished from the concept-level gold standard used for the paired evaluations. The training annotations identified textual spans, whereas the **evaluation assessed one unique morbidity concept per note**.

3.3.2 Annotation quality control

The initial Label Studio export contained **1,927 manually annotated morbidity spans**. A quality-control procedure was applied before the spans were converted into BIO labels. For **47 annotations**, the text value stored by Label Studio did not exactly match the corresponding substring at the recorded character positions. In these cases, the source discharge-summary text at the recorded positions was treated as authoritative.

Five exact duplicate spans were removed. A further **11 annotations** belonged to overlapping or nested groups that could not be represented unambiguously using a single BIO label sequence. In these cases, the longest clinically meaningful span was retained. Following these checks, 1,911 BIO-compatible morbidity spans remained for Bio_ClinicalBERT development.

These checks reduced avoidable label noise. Duplicate spans could cause the same phrase to be represented more than once, while incompatible overlapping spans could assign conflicting labels to the same tokens.

3.3.3 Operational definition of morbidity

For the final structured output, morbidity was defined as a disease or pathological condition affirmed for the patient. Both current and confirmed historical conditions were eligible for inclusion.

A disease mention was excluded from the primary morbidity output when it was:

- negated;
- uncertain, suspected or presented as a differential diagnosis;

- hypothetical or conditional;
- attributed to a family member or another person;
- included only in patient instructions or educational material; or
- present only in a non-diagnostic context, such as an allergy or medication list.

Medications, procedures, isolated anatomical structures and laboratory measurements were outside the scope of morbidity extraction. Symptoms and clinical findings were excluded unless they constituted a diagnosed pathological condition. Confirmed injuries, including fractures and haemorrhages, were retained when they satisfied the project's morbidity definition. Other clinical events could be retained in audit outputs but were not included in the primary disease-oriented evaluation.

This distinction was important because **detecting a disease term did not automatically establish that the patient had that disease**. For example, *pneumonia* in *no evidence of pneumonia* was a valid candidate morbidity mention, but it was not an eligible patient morbidity after contextual classification.

3.3.4 Refinement records and evaluation gold standards

The 60 records used during the second experimental phase supported manual output review and error analysis. Predictions from both pipelines were inspected to identify recurrent extraction, context, terminology and coding errors. Because these records influenced pipeline-development decisions, they were treated as development data and were excluded from both paired evaluations.

Two concept-level gold standards were constructed after pipeline development had been completed. The preliminary evaluation comprised 20 previously unseen discharge summaries containing **274 unique morbidity concepts**, while the expanded evaluation comprised 200 additional previously unseen discharge summaries containing **2,103** unique morbidity concepts. Eligible patient morbidities were recorded according to the operational definition in Section 3.3.3 and consolidated at concept level within each note.

Repeated textual occurrences of the same disease within a note were counted only once. Synonymous and abbreviated expressions were also consolidated when they represented the same clinical condition. For example, *DM2* and *type 2 diabetes mellitus* could be treated as a single morbidity concept when the clinical context supported the same diagnosis.

Concept-level consolidation prevented frequently repeated conditions from having a disproportionate effect on evaluation and aligned the gold standards with the intended system output: a concise list of distinct patient morbidities rather than every textual occurrence.

3.4 Data partitioning and leakage control

Table 3.1 summarises the experimental partitions and their roles.

Table 3.1 Experimental data partitions and their roles

Partition	Records	Purpose	Used for performance evaluation
Bio_ClinicalBERT training	70	Optimisation of model weights	No
Calibration	10	Checkpoint and confidence-threshold selection	No

Initial internal validation	20	Assessment of the initial transformer-centred pipeline	No
Pipeline refinement	60	Error analysis and iterative refinement of both pipelines	No
Preliminary paired evaluation	**20**	Initial evaluation of frozen pipelines	**Yes**
Expanded paired evaluation	**200**	Larger-scale evaluation of frozen pipelines	**Yes**

The **100-note** development corpus was divided into **70 training** notes, **10 calibration** notes and **20 initial internal-validatio**n notes. Before partitioning, the data-loading procedure verified that the note, subject and admission identifiers were unique and rejected duplicated identifiers. Each discharge summary in the development corpus therefore represented a different patient and hospital admission.

Partition assignment was based on *SHA-256 values* calculated from a fixed split seed together with each record's subject and note identifiers. The records were ranked by these values, making the split reproducible and independent of their original order in the source file. *Twenty records were first assigned to internal validation. The remaining records were ranked again using a calibration-specific version of the split key; 10 were assigned to calibration and the remaining 70 to training.* Partition sizes, calibration and validation identifiers, and source-data information were recorded in the experimental manifest. The training partition was reproducibly defined by the remaining records.

The training partition was used to optimise the Bio_ClinicalBERT model weights. A random seed of 42 was fixed for Python, NumPy and PyTorch operations to improve the reproducibility of stochastic model training. The calibration partition was used to select the model checkpoint and minimum entity-confidence threshold. The 20-note internal-validation partition was then used to assess the first complete version of the transformer-centred pipeline.

Because **the internal-validation results subsequently informed further development**, this partition was not treated as the final independent test set. The **60 additional refinement records** were also classified as development data because their outputs were reviewed and used to modify the two pipelines. The refinement procedure is described in Section 3.5.

The preliminary 20-record and expanded 200-record sets were reserved exclusively for evaluation after all model-selection and pipeline-refinement decisions had been completed. Both pipeline configurations were frozen before the preliminary evaluation and remained unchanged throughout the expanded evaluation. Neither evaluation set was used to retrain Bio_ClinicalBERT, select a checkpoint, adjust the confidence threshold, introduce contextual or terminology rules, or modify ICD-10 mapping procedures.

The expanded evaluation records were additionally kept patient-disjoint from the preliminary evaluation set. This ensured that the larger evaluation provided a new assessment of the same frozen configurations rather than an extension of the development process.

3.5 Iterative pipeline-refinement procedure

Following the initial internal assessment, pipeline refinement was conducted as an error-driven and iterative development process. **Both systems processed the 60 refinement records**, after which their structured outputs were manually reviewed. *Recurrent false-positive, false-*

negative, normalisation and coding errors were identified and grouped according to their likely causes. The relevant processing components were then modified, and the pipelines were reassessed on development data. This **review–refinement–reassessment cycle was repeated** during the development phase.

*For the **Bio_ClinicalBERT-centred pipeline**, refinement focused on:*

- contextual filtering of negated, uncertain and non-patient mentions;
- confidence-based selection of predicted morbidity spans;
- section-aware interpretation of clinical context;
- linking surface expressions to standard terminology concepts;
- consolidation of synonymous and repeated disease mentions;
- removal of non-disease concepts from the primary morbidity output.

These changes affected the complete transformer-centred pipeline rather than the Bio_ClinicalBERT model alone. Some improvements resulted from post-processing, contextual analysis, terminology linking and deduplication applied after neural span prediction.

*For the **cTAKES-centred pipeline**, refinement focused on:*

- additional assertion-processing safeguards;
- recognition and contextual disambiguation of clinical abbreviations;
- restriction to appropriate UMLS semantic types;
- removal of overly broad terminology matches;
- terminology supplementation for disease expressions missed by the initial configuration;
- selection between multiple UMLS concepts associated with the same mention;
- concept-level deduplication.

The shared **normalisation and coding stages were also reviewed**. Refinements addressed unresolved disease expressions, ambiguous SNOMED CT candidates, mappings requiring additional clinical detail and cases in which several potential ICD-10 families were available. When the terminology evidence supported a single reliable NHS ICD-10 family, it was recorded as the primary suggestion. When a unique family could not be determined safely, the pipeline retained a ranked list of up to five candidate three-character ICD-10 families for manual review rather than presenting any candidate as a definitive code. Candidate families were ranked using the level of support across the linked SNOMED CT concepts, lexical similarity to the official ICD-10 family title and the ordering information contained in the official mapping resource.

The refinements addressed recurring categories of error observed in development data rather than examples from either evaluation set. No preliminary or expanded evaluation record was used to introduce a new abbreviation, terminology rule, context pattern or mapping decision. Once the planned refinement experiments had been completed, **both pipeline configurations were frozen before the preliminary 20-record evaluation and remained unchanged throughout the subsequent 200-record expanded evaluation**.

Development-stage performance is reported separately in Chapter 5 to document the motivation for the refinement process. These development results are not interpreted as independent test performance because the records influenced subsequent pipeline decisions. The preliminary 20-record evaluation provided an initial paired assessment of the frozen systems, while the expanded 200-record evaluation provided the principal larger-scale estimate of their performance.

3.6 Evaluation design

3.6.1 Evaluation and matching rules

Both paired evaluations were conducted at the level of a unique morbidity concept within each discharge summary. A system prediction was compared with the gold standard for the same note. Matching was concept-based rather than restricted to identical character strings.

A prediction was accepted as a match when it represented the same clinical condition as a gold-standard morbidity, including an accepted synonym or abbreviation. Clinically distinct subtypes were not merged. For example, *type 1 and type 2 diabetes*, acute and chronic conditions, or *left*- and *right-sided disorders* were treated as separate concepts where the distinction was clinically relevant.

Repeated predictions representing the same morbidity within one note **were excluded as duplicates**. Consequently, each system could receive a maximum of one true positive for a particular morbidity concept in a particular note.

A prediction was classified as:

- **true positive** (TP) when it matched an eligible gold-standard morbidity;
- **false positive** (FP) when it did not correspond to an eligible gold-standard morbidity;
- **false negative** (FN) when a gold-standard morbidity was not identified by the system.

True negatives (TN) were not calculated because the number of all possible diseases not mentioned in a clinical record cannot be meaningfully enumerated.

3.6.2 Morbidity-extraction metrics

Precision *measures the proportion of correctly extracted morbidity concepts:*

$$Precision = \frac{TP}{TP + FP}.$$

Recall *measures the proportion of identified gold-standard morbidity concepts:*

$$Recall = \frac{TP}{TP + FN}.$$

*The **F1** score measures the balance between precision and recall:*

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}.$$

Precision, recall and F1 were calculated separately for the preliminary 20-record and expanded 200-record evaluations by aggregating TP, FP and FN counts across all records within each evaluation set. The two evaluation sets were not pooled when calculating the reported metrics.

Precision is particularly relevant when false diagnoses in a structured patient summary must be minimised.

Recall represents the completeness of the resulting morbidity list.

F1 provides a single summary measure that balances these two properties, although it does not indicate whether errors are primarily false positives or false negatives.

3.6.3 ICD-10 coding metrics

Coding performance was evaluated only for correctly extracted morbidity concepts. Three related measures were calculated.

***ICD-10 coding coverage** measured the proportion of true-positive morbidity concepts that received an ICD-10 suggestion:*

$$Coding\ coverage = \frac{\text{TP morbidities assigned a code}}{\text{all TP morbidities}}.$$

***ICD-10 coding-suggestion accuracy** measured the proportion of assigned code suggestions that were correct:*

$$Coding\ accuracy = \frac{\text{TP morbidities assigned the correct code}}{\text{TP morbidities assigned any code}}.$$

***Correct-code coverage** measured the proportion of all correctly extracted morbidity concepts that also received the correct code:*

$$Correct\text{-}code\ coverage = \frac{\text{TP morbidities assigned the correct code}}{\text{all TP morbidities}}.$$

ICD-10 coding metrics were calculated only for correctly extracted morbidity concepts for which a reliable reference coding judgement could be made. Where several candidate families were returned, the suggestion was accepted if the candidate set contained an appropriate three-character ICD-10 family.

These measures answer different questions. *Coding coverage* indicates how frequently the pipeline provides a suggestion. *Coding-suggestion accuracy* indicates how reliable the suggestions are when a code is provided. *Correct-code coverage* captures the end-to-end proportion of correctly extracted morbidities that are also correctly coded.

The evaluation was performed at the level of suggested three-character NHS ICD-10 5th Edition families. Where a pipeline returned more than one candidate family, the candidate set was marked as correct if it contained an acceptable three-character ICD-10 family.

Table 3.2 Interpretation of the evaluation metrics

Metric	Main question
Precision	How many extracted morbidities were correct?
Recall	How many gold-standard morbidities were found?
F1	What was the balance between precision and recall?

ICD-10 coding coverage	How many correctly extracted morbidities received a code?
ICD-10 coding-suggestion accuracy	How many assigned codes were correct?
ICD-10 correct-code coverage	How many correctly extracted morbidities received the correct code?

3.6.4 Evaluation implementation

Predictions from both pipelines were converted into a common structured review format containing the note identifier, extracted morbidity, normalised concept, terminology identifiers where available, and suggested NHS ICD-10 family. This common representation allowed outputs produced by the different technical environments to be reviewed and evaluated using the same concept-level criteria.

Manual review was used to determine whether each predicted morbidity corresponded to an eligible Gold-standard concept. Review decisions distinguished accepted concept matches, false-positive predictions and duplicate predictions excluded from scoring. Gold-standard morbidity concepts that were not represented among the accepted predictions for a system were subsequently counted as false negatives. Clinical equivalence was therefore determined through manual concept-level review rather than by exact string matching alone.

The preliminary 20-record evaluation used a structured workbook in which review decisions were recorded alongside the system predictions. The expanded 200-record evaluation used the same underlying morbidity definition, concept-matching policy, duplicate-exclusion rules and scoring criteria, but the review information was represented through explicit structured decision fields to support the substantially larger number of concepts. In both evaluations, the recorded manual decisions determined whether a morbidity prediction was accepted for scoring; the evaluation software aggregated these decisions but did not independently determine clinical equivalence.

ICD-10 assessment was performed only for coding-evaluable correctly extracted morbidity concepts. For each such concept, the review process recorded whether an ICD-10 suggestion had been produced and whether at least one suggested three-character NHS ICD-10 5th Edition family was considered acceptable. These decisions were then used to calculate ICD-10 coding coverage, coding-suggestion accuracy and correct-code coverage according to the definitions in Section 3.6.3. Where no reliable coding decision could be made, the concept was excluded from the coding-evaluable denominator rather than being treated automatically as an incorrect code.

The evaluation programs aggregated the reviewed decisions separately for the preliminary 20-record evaluation and the expanded 200-record evaluation. Precision, recall and F1 were calculated from the aggregated TP, FP and FN counts for each evaluation set, while the three coding measures were calculated from the corresponding coding-evaluable true-positive concepts. The two evaluation sets were not pooled when calculating the reported performance metrics.

Both pipelines processed the same records within each evaluation and were assessed against the same gold standard using identical morbidity definitions, concept-matching criteria and metric definitions. The comparisons were therefore descriptive and paired. No confidence intervals or formal statistical significance tests were calculated; consequently, numerical differences between the systems are interpreted as descriptive differences in operating behaviour rather than as evidence of statistically significant superiority.

4 System design and implementation

This chapter describes the technical implementation of two end-to-end pipelines for extracting morbidity entities. The first pipeline combined Bio_ClinicalBERT with contextual and terminology-based post-processing within EHRKit environment, while the second combined Apache cTAKES with additional assertion, terminology and mapping components.

4.1 Overall system architecture

Both pipelines received the same discharge-summary text and produced a comparable structured output. However, their internal extraction mechanisms differed. The transformer-centred pipeline used a fine-tuned Bio_ClinicalBERT model to identify morbidity spans from their linguistic context. The terminology-centred pipeline used cTAKES dictionary lookup, UMLS concepts and assertion attributes to identify candidate diseases.

After candidate extraction, both pipelines applied context processing, semantic filtering, concept normalisation, within-note deduplication and suggested NHS ICD-10 coding. The final outputs were converted into a common evaluation format containing one row for each unique morbidity concept identified within a note.

The **two implementations used different programming environments**. The Bio_ClinicalBERT-centred pipeline was implemented in Python within the EHRKit project environment, while the cTAKES-centred pipeline used Java and the Apache UIMA architecture. During final cTAKES execution, discharge-summary text and identifiers were supplied through CSV files, each note was processed as a temporary plain-text document, and the resulting annotations were exported as XMI for Java post-processing. Excel workbooks were used for structured result review and evaluation. These implementation differences did not affect the comparison because both pipelines processed the same clinical content and were assessed using identical rules.

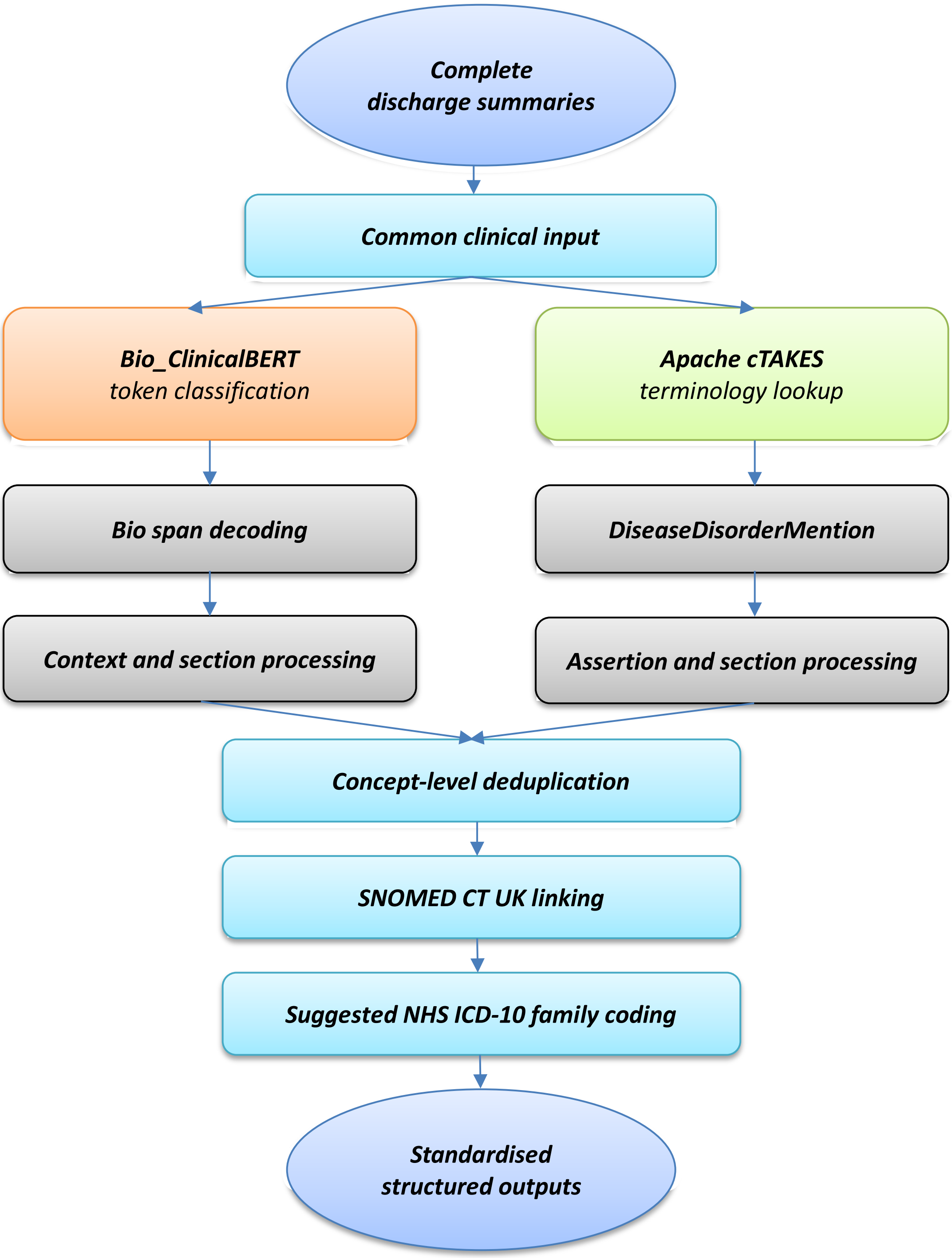

Figure 4.1 Architecture of the implemented morbidity-extraction and coding pipelines

4.2 Bio_ClinicalBERT-centred pipeline

4.2.1 Model architecture

Bio_ClinicalBERT was selected because its encoder had been pre-trained on clinical text, including MIMIC clinical notes [6]. The model was used within the EHRKit development environment [7] and adapted for token classification.

The implementation consisted of the pre-trained Bio_ClinicalBERT encoder, a dropout layer and a linear classification layer. For each input token, the classifier produced probabilities for three labels:

- O — outside a morbidity expression;
- B-MORBIDITY — the beginning of a morbidity expression;
- I-MORBIDITY — the continuation of a morbidity expression.

This design allowed the system to identify both single-word diseases and multi-word expressions such as chronic kidney disease or congestive heart failure.

Bio_ClinicalBERT was responsible primarily for predicting textual morbidity spans. Decisions concerning negation, uncertainty, patient experiencer, semantic type and coding were performed by subsequent components. The final Bio_ClinicalBERT result therefore represents the complete transformer-centred pipeline rather than the neural model in isolation.

4.2.2 Tokenisation and long-document processing

The manually annotated character spans were aligned with tokens produced by the Bio_ClinicalBERT WordPiece tokenizer. Only the first WordPiece of each original word received a supervised BIO label. Additional subword pieces were assigned an ignored value and did not contribute to the training loss. This prevented long medical terms that were divided into several WordPieces from receiving disproportionate weight during training.

Complete discharge summaries frequently exceeded the transformer input limit. Each note was therefore divided into overlapping windows of 256 tokens with a stride of 64 tokens. The overlap allowed a morbidity expression located near the end of one window to be represented again in the following window.

At inference, probability vectors produced for the same word in overlapping windows were averaged. The resulting word-level probabilities were decoded using a constrained Viterbi procedure. This permitted only valid BIO transitions: an I-MORBIDITY label could follow B-MORBIDITY or another I-MORBIDITY, but could not begin a new entity or directly follow O.

Consecutive B-MORBIDITY and I-MORBIDITY predictions were combined into complete character-level spans. Span boundaries were then adjusted to full-word boundaries, and predictions with implausibly short, excessively long or otherwise invalid boundaries were rejected.

4.2.3 Fine-tuning and model selection

The Bio_ClinicalBERT model was fine-tuned on the 70-note training partition described in Chapter 3. The training partition initially contained 1,173 manually annotated spans. Exact case-insensitive repetitions of an annotated phrase within the same training note were propagated as training-only weak supervision, increasing the number of training spans to 1,441. This operation was not applied to calibration, internal-validation or final evaluation records.

Table 4.1 presents the principal training parameters.

Table 4.1 Bio_ClinicalBERT training configuration

Parameter	Value
Base encoder	Bio_ClinicalBERT
Output labels	O, B-MORBIDITY, I-MORBIDITY
Maximum sequence length	256 tokens

Window stride	64 tokens
Batch size	2
Maximum epochs	8
Classifier learning rate	5×10^{-5}
Encoder learning rate	1×10^{-5}
Optimiser	AdamW
Weight decay	0.01
Linear warm-up	10% of training steps
Gradient clipping	1.0
Random seed	42
Selected epoch	7
Selected entity threshold	0.89

The model was trained for eight epochs and AdamW optimisation was used. The classifier used a learning rate five times larger than that of the encoder because its parameters were newly initialised for the morbidity task.

The checkpoint was selected using the 10-note calibration partition. Epoch 7 achieved the highest calibration selection score and was retained. Entity-confidence thresholds from 0.30 to 0.95 were then examined in increments of 0.01. The primary selection criterion was concept-level F1; ties were resolved by preferring higher recall and then the lower threshold. The selected minimum entity-confidence threshold was 0.89. *This operation was not applied to calibration, internal-validation or evaluation records.* **Both paired evaluations used the epoch-7 checkpoint trained on the 70-note training partition and selected using the 10-note calibration partition.** The additional records informed only the subsequent refinement of the complete processing workflow

An entity score was calculated from the token probabilities within its decoded span. Predictions below the selected threshold were excluded. The relatively high threshold made the model more conservative and contributed to the high precision observed in the expanded evaluation.

4.2.4 Contextual classification

Predicted morbidity spans were passed to medSpaCy ConText [17], which implements contextual processing derived from the ConText approach [11]. The component supplied indicators for properties such as negation, uncertainty, historical status and experiencer.

Additional section-aware rules were applied because complete discharge summaries contain headings, lists and hard-line breaks that can cause contextual triggers to extend beyond their intended scope. Modifier cues were therefore constrained to the current clinical section and local clause. For example, denies in a Review of Systems section was not allowed to negate a diagnosis appearing later in the discharge summary.

Each candidate mention was assigned one of the following contextual classes:

- confirmed current;
- confirmed historical;

- suspected or possible;
- negated;
- family history;
- instruction or education;
- non-diagnostic context.

Only confirmed current and confirmed historical conditions attributed to the patient were eligible for the primary morbidity output. Negated, uncertain, hypothetical, family-related and instruction-only mentions remained available in the audit output but were excluded from the evaluated disease list.

This stage is important when interpreting the results. Bio_ClinicalBERT proposed the textual spans, but the complete pipeline—including medSpaCy ConText, section rules and semantic filters—determined whether a predicted span represented an eligible patient morbidity.

4.2.5 Semantic filtering and post-processing

A terminology-supported semantic profile was constructed for each contextually eligible mention. UMLS semantic types and SNOMED CT semantic tags were used to distinguish diseases from symptoms, findings, injuries, procedures, medications and anatomical structures [4].

Only concepts satisfying the operational morbidity definition in Section 3.3.3 were included in the primary morbidity output. Symptoms and clinical findings were excluded unless they represented a diagnosed pathological condition, while confirmed injuries were retained when they satisfied the same eligibility criteria. Other ineligible findings and clinical events remained available in the audit output but were not included in the primary evaluation.

Repeated predictions were deduplicated within each note. Where a SNOMED CT identifier was available, it formed the principal deduplication key. Otherwise, the normalised morbidity expression was used. When several mentions represented the same concept, the pipeline retained a representative row using contextual status, clinical-section priority, confidence and mention length while preserving the number of merged occurrences for audit purposes.

4.3 cTAKES-centred pipeline

4.3.1 Pipeline execution and XMI processing

The terminology-centred implementation used Apache cTAKES 6.0.0 [5]. During both paired evaluations, the Java orchestration program used a CSV file containing the note, subject and admission identifiers together with the complete discharge-summary text. Each note was converted into a temporary plain-text file and processed using the local cTAKES clinical pipeline and UMLS terminology configuration. The resulting cTAKES annotations were exported as XMI files for subsequent parsing and post-processing by the Java pipeline.

cTAKES produced an XMI representation of each processed note. The Java implementation parsed these XMI files and extracted DiseaseDisorderMention annotations together with their character positions, preferred terms, UMLS Concept Unique Identifiers and semantic types [4].

The local pipeline also provided assertion fields including:

- polarity;
- uncertainty;
- conditionality;

- generic status;
- historical status;
- subject or experiencer.

The resulting cTAKES annotations formed the primary candidates for the terminology-centred branch.

4.3.2 Disease-concept selection

A cTAKES mention could be associated with several UMLS concepts. Candidate concepts were therefore scored according to their semantic type, lexical agreement with the extracted phrase, preferred term, terminology source and availability of an official NHS mapping.

*The primary disease-oriented **UMLS semantic types** retained by the implementation were:*

- T019 — Congenital Abnormality;
- T020 — Acquired Abnormality;
- T037 — Injury or Poisoning;
- T047 — Disease or Syndrome;
- T048 — Mental or Behavioural Dysfunction;
- T191 — Neoplastic Process.

These semantic types were used to retain candidate disease and pathological-condition concepts for further processing; they did not guarantee automatic inclusion in the final output. In particular, T037 concepts were retained only when the extracted injury represented an eligible confirmed patient condition under the project's morbidity definition.

Concepts corresponding only to findings, anatomical structures, procedures or medications were excluded. Generic expressions such as disease, problem, mass or history were also rejected when they did not identify a sufficiently specific morbidity.

4.3.3 Assertion and context processing

The Java pipeline interpreted the cTAKES assertion attributes together with additional section- and scope-aware safeguards.

A mention was excluded when it was:

- assigned negative polarity;
- uncertain or conditional;
- hypothetical or contained in patient instructions;
- attributed to a non-patient subject;
- located in a family-history section;
- generic rather than diagnostic;
- located only in an excluded section such as medications, allergies or laboratory data.

Historical patient diseases were retained, as required by the morbidity definition. Additional bidirectional context patterns handled constructions in which negation or uncertainty appeared either before or after the disease expression, such as no evidence of pneumonia and pneumonia is not present.

Consequently, cTAKES did process clinical context, but it did so through assertion attributes and explicit rules rather than through the contextual representation used by a transformer.

4.3.4 Terminology and abbreviation supplements

The final cTAKES-centred pipeline was not limited to the unmodified cTAKES output. A supplementary abbreviation pass recognised common clinical forms such as *HTN, COPD, CKD, DM2* and *NSTEMI*. Context-sensitive rules were applied to ambiguous abbreviations before an expansion was accepted.

A further terminology pass used exact disease expressions from UMLS 2026AA when cTAKES had produced no overlapping disease annotation. This was intended to improve coverage for valid clinical expressions missing from the local cTAKES lookup configuration. The supplement used terminology-based disease expressions rather than a table derived from either evaluation set.

A limited set of spelling normalisations was also applied before concept matching. These additions increased terminology coverage but mean that the final results should be attributed to the complete cTAKES-centred pipeline, not to standard cTAKES alone.

4.3.5 cTAKES deduplication

Deduplication was performed in two stages. First, exact normalised textual repetitions were consolidated within each note. If the same phrase was produced by more than one extraction component, priority was given to the primary cTAKES annotation, followed by the abbreviation supplement and then the additional terminology pass.

After terminology mapping, rows were grouped using SNOMED CT identifiers, UMLS CUIs and compatible normalised concepts. Compatibility guards prevented clinically distinct conditions from being merged.

These guards preserved distinctions including:

- type 1 versus type 2 diabetes;
- acute versus chronic disease;
- left- versus right-sided conditions;
- STEMI versus NSTEMI;
- traumatic versus non-traumatic disease;
- subdural versus epidural conditions.

The representative row was selected using mapping provenance, availability of terminology identifiers, assertion status and extraction source. Alternative surface expressions and the number of merged occurrences were retained for auditing.

4.4 Concept normalisation and SNOMED CT UK linking

Concept normalisation converted different textual expressions representing the same clinical condition into a comparable form. Basic text normalisation standardised case, whitespace and punctuation. Roman numerals were converted only when they occurred in explicit constructions such as disease type, stage, grade or class.

The transformer-centred pipeline attempted to link extracted expressions to active SNOMED CT UK descriptions. Exact normalised SNOMED synonyms and UMLS aliases linked to active UK concepts were preferred. If no exact match was available, a conservative lexical comparison was applied. A lexical match was accepted only when it achieved a sufficiently high score, differed clearly from alternative candidates and preserved clinically important modifiers such as type, stage, anatomical site, laterality and acuity.

The cTAKES-centred pipeline began with its UMLS CUI and preferred term. Where possible, these were linked to a corresponding SNOMED CT UK concept. Additional lexical matching was used when the cTAKES CUI did not provide a unique UK SNOMED CT identifier.

Uncertain concept matches were retained for review rather than silently promoted to confirmed concepts. This was important because an incorrect terminology link could subsequently generate an incorrect ICD-10 suggestion.

4.5 Suggested NHS ICD-10 coding

Eligible concepts were mapped using the official SNOMED CT UK to ICD-10 complex-map reference set and the NHS ICD-10 5th Edition codes and titles [2], [3], [16]. The mapping process did not use ICD-10-CM codes or MIMIC diagnosis-code fields as substitutes for NHS ICD-10.

The pipeline evaluated the mapping evidence associated with each candidate SNOMED CT concept. When the evidence supported one reliable three-character NHS ICD-10 family, that family was recorded as the primary suggestion. Conditional mappings, multiple map blocks and mappings requiring clinical details unavailable in the extracted text were marked for review.

When a unique family could not be determined safely, the pipeline retained a ranked list of up to five candidate ICD-10 families. Candidate families were ranked using:

- support across the linked SNOMED CT concepts;
- lexical similarity to the official ICD-10 family title;
- the priority, block and group ordering provided by the official mapping resource.

Candidate lists were provided for manual review and were not presented as definitive coding decisions. The output should therefore be interpreted as coding support rather than automatic replacement of a qualified clinical coder.

4.6 Structured output and audit information

Both pipelines were converted into a common logical output. Table 4.2 presents the principal fields used in the comparison.

Table 4.2 Common structured output fields

Field	Description
note_id	Discharge-summary identifier
subject_id	Patient identifier
hadm_id	Hospital-admission identifier
morbidity	Extracted textual expression
normalised_morbidity	Standardised expression
clinical_status	Current, historical, uncertain, negated or excluded status
UMLS_CUI	UMLS concept identifier, where available
SNOMED_CT_ID	SNOMED CT UK identifier, where available
ICD_10_suggested	Suggested ICD-10 family or ranked candidates
source_system	Bio_ClinicalBERT-centred or cTAKES-centred pipeline

needs_review	Indicator of unresolved or ambiguous mapping

The Bio_ClinicalBERT workbook separated confirmed diseases, confirmed non-disease conditions, all candidate mentions, excluded mentions and processing statistics. The cTAKES workbook similarly retained accepted morbidity concepts, excluded assertion cases, terminology provenance, mapping candidates and pipeline statistics.

This audit information made it possible to trace a final suggestion back to its textual evidence, contextual decision, terminology link and mapping source. The common output format was then used to construct the manual evaluation workbook described in Chapter 3.

5 Results

This chapter reports the development-stage assessment and the preliminary and expanded paired evaluation results for the two implemented pipelines. In the figures, the label EHRKit refers to the complete Bio_ClinicalBERT-centred pipeline rather than to the neural model alone.

5.1 Initial Bio_ClinicalBERT assessment

The initial model-development process selected epoch 7 and an entity-confidence threshold of 0.89. Table 5.1 presents the saved calibration and internal-validation results at the unique-concept level.

Table 5.1 Initial Bio_ClinicalBERT development-stage results

Stage	TP	FP	FN	Precision	Recall	F1
Calibration at threshold 0.89	78	33	46	70.27%	62.9%	66.38%
Initial internal validation	130	81	66	61.61%	66.33%	63.88%

The initial internal-validation F1 score of 63.88% did not meet the pre-specified 80% internal quality gate. The optional refit across all 100 annotated notes was therefore not performed at that stage. Instead, the observed errors motivated the iterative refinement procedure described in Sections 3.5 and 4.2, 4.3, 4.4, 4.5.

The development-stage results should not be compared directly with the paired evaluation results as if they represented performance before and after a single controlled intervention. They were calculated on different records and reflected different stages of the processing workflow. They nevertheless document why further contextual, terminology, deduplication and coding refinements were necessary.

The cTAKES-centred pipeline was also refined using the separate development records. The two pipelines should therefore be compared directly only within the shared preliminary and expanded evaluation sets.

5.2 Evaluation dataset

Following pipeline development and refinement, both configurations were frozen. A preliminary paired evaluation was first conducted on 20 previously unseen, patient-disjoint discharge summaries containing 274 gold-standard morbidity concepts. The evaluation was subsequently expanded to 200 previously unseen discharge summaries, producing a substantially larger gold standard of 2,103 morbidity concepts. No model training, threshold selection, terminology refinement or rule modification was performed using either evaluation set, and the same morbidity definition, concept-matching rules and scoring procedure were retained throughout.

Table 5.2 Evaluation datasets

Evaluation	Records	Gold morbidity concepts	Role
Preliminary evaluation	20	**274**	Initial paired assessment
Expanded evaluation	200	**2,103**	Larger-scale evaluation

In the preliminary 20-record evaluation, the **Bio_ClinicalBERT**-centred pipeline produced **213** unique morbidity predictions, comprising **194** true positives and **19** false positives, while **80** gold-standard concepts were missed. The **cTAKES**-centred pipeline produced **266** predictions, comprising **216** true positives and **50** false positives, with **58** false negatives.

In the expanded 200-record evaluation, the same operating pattern remained evident. The **Bio_ClinicalBERT**-centred pipeline produced **1,530** unique morbidity predictions, of which **1,378** were true positives and **152** were false positives; **725** gold-standard morbidities were missed. The **cTAKES**-centred pipeline produced **2,220** predictions, including **1,590** true positives and **630** false positives, while **513** gold-standard morbidities were missed.

The larger evaluation therefore preserved the distinction observed in the preliminary experiment. *Bio_ClinicalBERT continued to return a smaller and more conservative set of morbidity predictions, whereas cTAKES identified a larger number of gold-standard conditions but also generated substantially more false-positive concepts*. The corresponding precision, recall and F1 results are presented in Section 5.3.

5.3 Morbidity-extraction results

Table 5.3 reports precision, recall and F1 for both the preliminary 20-record evaluation and the expanded 200-record evaluation, while *Figure 5.1* provides a visual comparison of the two systems across both evaluation sizes.

Table 5.3 Morbidity-extraction performance across the two evaluations

System	Evaluation	Precision	Recall	F1
Bio_ClinicalBERT	20 records	91.08%	70.8%	79.67%
Bio_ClinicalBERT	**200 records**	**90.07%**	**65.53%**	**75.86%**
cTAKES	20 records	81.2%	78.83%	80%
cTAKES	**200 records**	**71.62%**	**75.61%**	**73.56%**

The preliminary evaluation already indicated contrasting operating profiles. Bio_ClinicalBERT achieved substantially higher precision (91.08%) but lower recall (70.80%), whereas cTAKES achieved lower precision (81.20%) but higher recall (78.83%). Their F1 scores were almost identical at this stage (79.67% and 80.00%, respectively), indicating that the two pipelines reached a similar overall balance through different precision–recall trade-offs.

The expanded 200-record evaluation preserved the same distinction between the systems. Bio_ClinicalBERT achieved a precision of 90.07%, recall of 65.53% and F1 of 75.86%, while cTAKES achieved 71.62% precision, 75.61% recall and 73.56% F1. On the larger evaluation set, Bio_ClinicalBERT therefore had 18.45 percentage points higher precision, whereas cTAKES had 10.08 percentage points higher recall. Bio_ClinicalBERT achieved the higher F1 score by 2.30 percentage points.

Compared with the preliminary evaluation, Bio_ClinicalBERT precision remained relatively stable, whereas the largest change was the reduction in cTAKES precision; recall decreased more moderately for both systems.

Overall, the expanded evaluation strengthened the interpretation established by the preliminary experiment: **Bio_ClinicalBERT remained the more selective and precision-oriented pipeline, whereas cTAKES continued to retrieve a broader proportion of the gold-standard morbidities**. The larger sample therefore provided a more robust estimate of absolute performance while retaining the same underlying precision–recall trade-off.

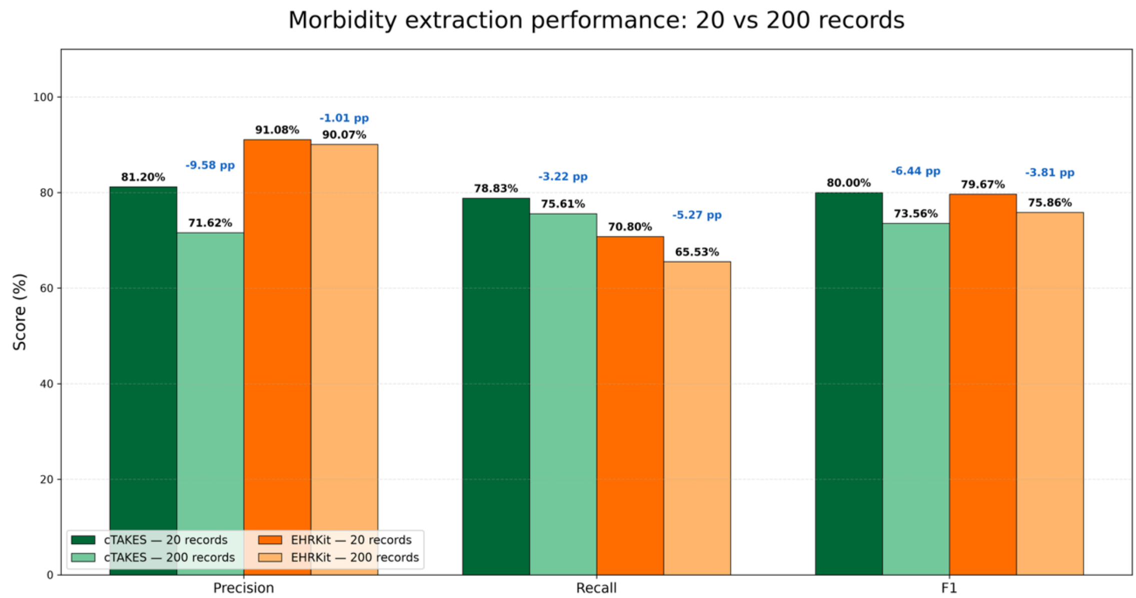

Figure 5.1 Morbidity-extraction performance in the preliminary 20-record and expanded 200-record evaluations

5.4 ICD-10 coding results

Table 5.4 compares the ICD-10 coding metrics obtained in the preliminary 20-record evaluation and the expanded 200-record evaluation, while Figure 5.2 provides a visual comparison of coding coverage, correct-code coverage and coding-suggestion accuracy across both evaluation sizes.

Table 5.4 Suggested NHS ICD-10 coding performance across the two evaluations

System	Evaluation	Coding coverage	Correct-code coverage	Coding-suggestion accuracy
Bio_ClinicalBERT	20 records	90.21%	85.05%	94.29%
Bio_ClinicalBERT	**200 records**	**93.68%**	**83.28%**	**88.91%**
cTAKES	20 records	100%	95.83%	95.83%
cTAKES	**200 records**	**94.52%**	**87.41%**	**92.47%**

The preliminary evaluation indicated strong coding performance for both pipelines, with an advantage for cTAKES. cTAKES achieved complete ICD-10 coding coverage (100.00%) and 95.83% for both correct-code coverage and coding-suggestion accuracy. Bio_ClinicalBERT achieved 90.21% coding coverage, 85.05% correct-code coverage and 94.29% coding-suggestion accuracy.

The expanded 200-record evaluation preserved the coding advantage of cTAKES, although the difference in ICD-10 coding coverage became small. cTAKES achieved 94.52% coding coverage, 87.41% correct-code coverage and 92.47% coding-suggestion accuracy. Bio_ClinicalBERT achieved 93.68%, 83.28% and 88.91%, respectively. Thus, *on the larger evaluation set, cTAKES* *exceeded Bio_ClinicalBERT by 0.84 percentage points in coding coverage, 4.13 percentage points in correct-code coverage and 3.56 percentage points in coding-suggestion accuracy.*

ICD-10 coding coverage converged on the larger evaluation and was high for both pipelines, while cTAKES retained clearer advantages in coding-suggestion accuracy and correct-code coverage.

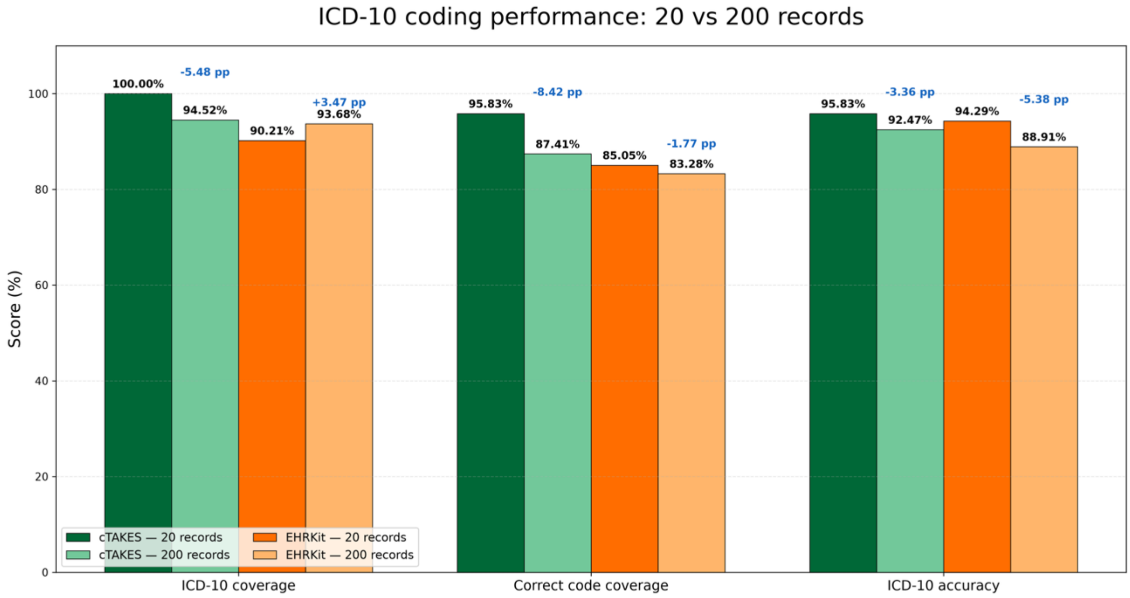

Figure 5.2 Suggested NHS ICD-10 coding performance across the two evaluations

5.5 Complementarity and error analysis

5.5.1 Cross-system complementarity

To examine whether the two pipelines identified the same or different gold-standard morbidity concepts, their true-positive detections were compared at concept level across the expanded 200-record evaluation. Table 5.5 shows the proportion of gold-standard morbidities detected by both systems, detected uniquely by either pipeline, or missed by both.

Table 5.5 Cross-system overlap of gold-standard morbidity detection in the expanded 200-record evaluation

Outcome	Gold concepts	%
Detected by both systems	1,126	53.54%
Detected only by Bio_ClinicalBERT	252	11.98%
Detected only by cTAKES	464	22.06%
Missed by both systems	261	12.41%
Detected by at least one system	**1,842**	**87.59%**
Total Gold concepts	2,103	100%

The overlap analysis demonstrated substantial complementarity between the two pipelines. Overall, **87.59% of the 2,103 gold-standard morbidity concepts were detected by at least one system**, while Table 5.5 shows that each pipeline also recovered concepts missed by the other. This does not establish the performance of a hybrid system, because simply combining the predictions would also combine false positives. However, the pattern provides quantitative motivation for the hybrid architecture considered in Chapter 6.

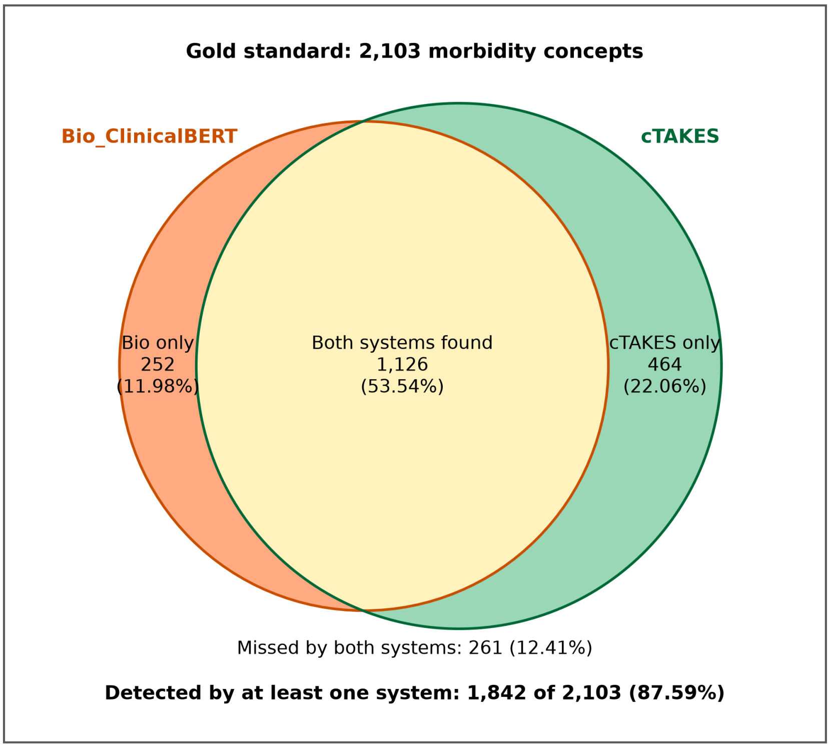

Figure 5.3 Cross-system complementarity in gold standard morbidity detection in the expanded 200-record evaluation

5.5.2 Bio_ClinicalBERT-centred pipeline

The principal quantitative limitation of the Bio_ClinicalBERT-centred pipeline was recall. In the expanded evaluation, it **missed 725 of the 2,103** gold-standard morbidity concepts.

Development-stage error analysis and subsequent output review indicated several possible sources of missed morbidities:

- uncommon diseases insufficiently represented in the annotated training data;
- abbreviations and spelling variants not learned or normalised successfully;
- predicted spans falling below the 0.89 confidence threshold;
- incomplete or incorrect span boundaries;
- information divided across long-document windows;
- eligible conditions removed by conservative contextual or semantic filtering.

The high confidence threshold and strict post-processing rules reduced false-positive predictions but could also reject valid low-confidence conditions. This helps explain why Bio_ClinicalBERT retained **high precision (90.07%) but lower recall (65.53%)** in the expanded evaluation.

The pipeline produced **152 false-positive concepts** across the 200 records. Although this absolute count is larger than in the preliminary evaluation because substantially more records were assessed, the resulting precision remained very high and changed only slightly from 91.08% to

90.07%. Examples of false-positive error types identified during development and output review included contextually ambiguous mentions and clinical findings incorrectly interpreted as eligible morbidities.

5.5.3 cTAKES-centred pipeline

The cTAKES-centred pipeline **missed 513 of the 2,103** gold-standard morbidity concepts and therefore retained higher recall than Bio_ClinicalBERT.

Its greater coverage was supported by its broad terminology resources, abbreviation handling and supplementary terminology matching. However, *the same broad coverage also introduced several potential sources of false-positive predictions:*

- dictionary matches that were medically valid concepts but not eligible patient morbidities;
- ambiguous abbreviations assigned an incorrect disease meaning;
- broad UMLS disease categories that did not match the intended gold-standard concept;
- assertion modifiers extending beyond their intended scope;
- mentions in historical, instructional or non-diagnostic contexts being interpreted incorrectly;
- several candidate concepts being available for the same surface expression.

The cTAKES-centred pipeline produced **630 false-positive concepts** in the expanded evaluation, resulting in a precision of **71.62%**, compared with **90.07%** for Bio_ClinicalBERT. At the same time, its recall remained higher at **75.61%**, compared with **65.53%** for Bio_ClinicalBERT. The custom assertion and semantic safeguards therefore reduced but did not eliminate the false positives associated with broad terminology-driven extraction.

Overall, the expanded evaluation reinforced the same trade-off observed in the preliminary experiment: **cTAKES provided broader morbidity coverage, whereas Bio_ClinicalBERT applied a more selective extraction strategy**.

5.5.4 Terminology and ICD-10 mapping errors

In the expanded evaluation, **87 of the 1,376 coding-evaluable correctly extracted morbidities from the Bio_ClinicalBERT-centred pipeline did not receive an ICD-10 suggestion**, resulting in ICD-10 coding coverage of 93.68%. For cTAKES, **87 of 1,588 coding-evaluable correctly extracted morbidities did not receive a suggestion,** resulting in slightly higher coverage of 94.52%.

Among the morbidities for which a code was assigned, Bio_ClinicalBERT produced **1,146 correct suggestions from 1,289 coded concepts**, corresponding to a coding-suggestion accuracy of 88.91%. cTAKES produced **1,388 correct suggestions from 1,501 coded concepts**, corresponding to a higher coding-suggestion accuracy of 92.47%. Thus, the expanded evaluation showed that the difference between the systems was not primarily whether a code could be generated, since ICD-10 coding coverage was high and similar for both pipelines, but rather the proportion of assigned suggestions that were correct.

Missing or incorrect mappings identified during terminology and output review could occur when:

- an extracted phrase could not be linked confidently to an active SNOMED CT UK concept;

- several SNOMED CT concepts matched the same expression;
- the clinical text lacked information about disease type, stage, cause, anatomical site or complication;
- the official map contained conditional rules or multiple map blocks;
- the terminology concept and ICD-10 family represented different levels of clinical detail.

The ranked candidate lists preserved possible mappings for review, but the presence of a candidate did not necessarily mean that it was accepted as a correct code during evaluation.

Because the evaluation was conducted at the level of three-character NHS ICD-10 5th Edition families, the reported coding-suggestion accuracy does not establish accuracy for complete four- or five-character coding. More specific coding would require additional clinical information and validation by a qualified clinical coder.

5.6 Results summary

Neither pipeline performed best across all evaluated measures. In the expanded 200-record evaluation, the **Bio_ClinicalBERT-centred pipeline achieved substantially higher precision and a modestly higher F1 score**, whereas the **cTAKES-centred pipeline achieved higher recall and stronger ICD-10 coding performance**, including higher coding-suggestion accuracy and correct-code coverage.

The expanded evaluation therefore **reinforced the principal pattern observed in the preliminary experiment rather than changing its interpretation**. Bio_ClinicalBERT remained the more selective and precision-oriented approach, while cTAKES continued to capture a broader proportion of the gold-standard morbidities. The substantial number of concepts identified by only one pipeline further demonstrated their complementary behaviour and provided quantitative motivation for the hybrid architecture considered in Chapter 6.

5.7 Practical reflection on EHRKit and cTAKES

Learnability: From my experience during this project, cTAKES was considerably easier to learn and begin using than the Bio_ClinicalBERT/EHRKit pipeline. Once the required cTAKES environment and terminology resources had been configured, its processing stages and outputs were relatively straightforward to understand and inspect. In contrast, considerably more time was required to establish the Bio_ClinicalBERT/EHRKit environment and understand the complete transformer-based workflow. This included preparing annotated data, converting annotations into the required training format, fine-tuning the model, selecting a checkpoint and confidence threshold, and resolving implementation and dependency-related problems. As a result, the initial learning curve was substantially steeper for the transformer-centred pipeline.

Ease of use: cTAKES was also easier to use during the iterative development process. Its terminology-centred architecture made it possible to inspect recognised concepts, UMLS identifiers and assertion information directly, which was useful when tracing errors and modifying individual processing components. Many changes could be made through terminology supplementation, semantic filtering or assertion rules without retraining the underlying system. By comparison, the Bio_ClinicalBERT/EHRKit pipeline involved a larger number of interdependent stages, including model inference, confidence-based filtering, contextual processing, terminology normalisation and deduplication. Model-related errors were also less straightforward to correct individually, since changing a particular prediction could require additional training data or changes to the surrounding post-processing rather than a simple terminology or rule adjustment.

Breadth of functionality and features: cTAKES provided a broad set of immediately useful clinical NLP functions, particularly terminology lookup, UMLS concept linking and assertion-related information. These components were well suited to the terminology-centred requirements of this project and contributed to the relatively rapid construction of a complete extraction and coding workflow. Bio_ClinicalBERT itself provided contextual span recognition rather than the same breadth of terminology functionality, meaning that additional components were required for context handling, normalisation and ICD-10 mapping. However, the transformer-centred approach appeared to offer greater potential for task-specific adaptation. Because Bio_ClinicalBERT can be further fine-tuned on additional annotated clinical data, its behaviour is not limited to a fixed terminology or set of manually defined rules. A larger and more diverse training corpus could therefore potentially improve its recognition of uncommon conditions, abbreviations and variable clinical expressions.

Overall reflection: In practical terms, cTAKES was the easier and more immediately productive toolkit for this project, while the Bio_ClinicalBERT/EHRKit pipeline required substantially more development effort. However, ease of use did not necessarily correspond to long-term potential. The transformer-centred approach appeared more adaptable to further task-specific training, whereas cTAKES provided stronger ready-made terminology and concept-processing functionality. This practical experience was consistent with the quantitative evaluation, in which the two approaches demonstrated different strengths rather than one consistently dominating the other. Taken together, these observations suggest that a particularly promising future direction would be a hybrid pipeline combining the terminology coverage, interpretability and practical usability of cTAKES with the contextual modelling and trainability of Bio_ClinicalBERT.

6 Evaluation and Conclusions

This chapter evaluates the extent to which the project answered its research questions and achieved its objectives. It also considers the contribution of the findings in relation to existing work, identifies the principal limitations and proposes directions for further development.

6.1 Summary

This project developed and evaluated two end-to-end pipelines for extracting patient morbidities from complete MIMIC-IV discharge summaries and assigning suggested NHS ICD-10 5th Edition families. The first pipeline combined a fine-tuned Bio_ClinicalBERT sequence tagger with contextual, semantic and terminology-based post-processing, while the second combined Apache cTAKES with UMLS concept extraction, assertion processing, terminology supplementation and UK-specific mapping.

The experimental design separated model development, pipeline refinement and evaluation using patient-disjoint records. Bio_ClinicalBERT was developed using 100 manually annotated discharge summaries, while a further 60 records supported iterative refinement of both complete pipelines. After both configurations were frozen, a preliminary paired evaluation on 20 previously unseen records containing 274 gold-standard morbidity concepts was followed by an expanded evaluation on 200 previously unseen records containing 2,103 gold-standard concepts. The same morbidity definition, concept-matching rules and scoring procedure were retained across both evaluations.

The expanded evaluation confirmed the contrasting operating profiles observed in the preliminary experiment. Bio_ClinicalBERT achieved **90.07% precision, 65.53% recall and 75.86% F1**, whereas cTAKES achieved **71.62% precision, 75.61% recall and 73.56% F1**. Bio_ClinicalBERT therefore remained the more selective and precision-oriented pipeline, while cTAKES identified a broader proportion of the gold-standard morbidities. Neither approach was uniformly superior across the extraction measures.

For suggested ICD-10 assignment, cTAKES achieved stronger overall coding performance, with **92.47% coding-suggestion accuracy** and **87.41% correct-code coverage**, compared with **88.91%** and **83.28%** for Bio_ClinicalBERT, respectively. ICD-10 coding coverage was high and similar for both pipelines (**94.52% for cTAKES and 93.68% for Bio_ClinicalBERT**). Cross-system overlap analysis additionally demonstrated substantial complementarity between the two extraction profiles.

6.2 Evaluation of the project

6.2.1 Evaluation against research questions

1. ***The first research question*** *asked how accurately the Bio_ClinicalBERT-centred and cTAKES-centred pipelines could extract unique morbidity concepts from complete MIMIC-IV discharge summaries.*

The expanded evaluation demonstrated a clear precision–recall trade-off between the two approaches. Bio_ClinicalBERT achieved substantially higher precision (90.07%), whereas cTAKES achieved higher recall (75.61%). Bio_ClinicalBERT also achieved a modestly higher F1 score (75.86% vs 73.56%), a difference of 2.30 percentage points. However, these differences are descriptive rather than statistically significant, as no formal significance testing was performed. The direction of the trade-off was consistent across both the preliminary and expanded evaluations: Bio_ClinicalBERT remained more precision-oriented, while cTAKES consistently retrieved a larger proportion of the Gold-standard morbidities. Neither pipeline was therefore superior across all extraction dimensions.

2. ***The second research question*** *asked how effectively the extracted morbidities could be assigned suggested NHS ICD-10 5th Edition families.*

ICD-10 coding coverage was high and similar for both pipelines, reaching 93.68% for Bio_ClinicalBERT and 94.52% for cTAKES. However, cTAKES achieved higher coding accuracy (92.47% vs 88.91%) and higher correct-code coverage (87.41% vs 83.28%). These results indicate that, although both pipelines were able to produce ICD-10 suggestions for a high proportion of coding-evaluable extracted morbidities, the cTAKES-centred pipeline demonstrated stronger terminology-based coding performance in the expanded evaluation.

3. ***The third research question*** *examined the complementary strengths, limitations and error patterns of the transformer-based and terminology-driven approaches.*

The principal **strength of the Bio_ClinicalBERT-centred pipeline was precision**. Its learned representation, confidence threshold and contextual post-processing enabled it to reject many inappropriate disease candidates. Its **principal limitation was recall:** uncommon conditions, abbreviations and low-confidence mentions were more likely to be missed.

The principal **strength of the cTAKES-centred pipeline was terminology coverage**. Its UMLS resources, abbreviation handling and supplementary terminology processing allowed it to identify more gold-standard morbidities and achieve stronger terminology-based coding performance. Its principal **limitation was the number of false positives** generated by broad dictionary matches, ambiguous abbreviations and imperfect assertion or semantic filtering.

The cross-system analysis showed that the two pipelines made substantially different errors, with each system recovering morbidity concepts missed by the other. This supports the interpretation that their strengths are complementary rather than interchangeable. The third research question was therefore answered through the identification of a clear **precision–recall trade-off** and distinct error profiles, which together motivate the hybrid architecture proposed later in this chapter.

6.2.2 Evaluation against the project objectives

Table 6.1 evaluates the work against the objectives defined in Section 1.3.

Table 6.1 Evaluation of the project objectives

Objective	Evidence of achievement	Outcome
Prepare MIMIC-IV discharge summaries and create a manually annotated gold standards	Complete discharge summaries were selected and separated at patient level; the expanded evaluation gold standard contained **2,103 unique morbidity concepts across 200 previously unseen records**	**Achieved**
Develop and fine-tune a Bio_ClinicalBERT morbidity-extraction pipeline within EHRKit	A BIO token-classification model was trained and calibrated, then integrated with contextual, terminology, deduplication and coding components	**Achieved**
Configure cTAKES for morbidity extraction from the same type of records	cTAKES 6.0.0 was integrated with UMLS extraction, assertion processing, terminology supplementation, semantic filtering and UK-specific mapping	**Achieved**

Apply context filtering and within-record deduplication	Both pipelines excluded ineligible contextual mentions and consolidated repeated or synonymous morbidity concepts within each record	**Achieved**
Normalise morbidities and map them to suggested NHS ICD-10 families	UMLS and SNOMED CT UK resources were linked to NHS ICD-10 mapping resources, with candidate suggestions retained where a unique mapping could not be established	**Achieved with limitations**
Compare the pipelines using precision, recall and F1	Both frozen pipelines were evaluated under identical rules on the same preliminary and expanded patient-disjoint evaluation records	**Achieved**
Evaluate coding coverage and accuracy and analyse system strengths and limitations	Coding coverage, correct-code coverage and coding-suggestion accuracy were calculated, together with comparative error and cross-system complementarity analysis	**Achieved**

The objectives were therefore substantially achieved. The qualification applied to terminology normalisation and coding reflects the fact that some correctly extracted morbidities did not receive a reliable ICD-10 coding suggestion.

The project followed a controlled and reproducible evaluation design. As described in Chapter 3, patient-disjoint development, refinement and evaluation data were kept separate, and neither the preliminary nor the expanded evaluation records were used for model selection, threshold calibration, pipeline refinement or rule development. Both frozen pipelines were evaluated on the same clinical texts within each evaluation and assessed against the same gold standard using consistent morbidity definitions, concept-matching rules and scoring procedures. The expansion from 20 to 200 previously unseen discharge summaries provided a substantially larger basis for assessing performance while preserving the principal pattern observed in the preliminary experiment.

6.3 Comparison with Existing Work

The results are consistent with previous research showing that clinical concept-recognition performance depends strongly on the dataset, entity definition, terminology coverage and evaluation method. Lossio-Ventura et al. reported substantial variation among clinical concept-recognition systems across exact and inexact matching, negation, abbreviations, ambiguity and misspellings [8]. *The present study similarly found that no system was optimal across all evaluation measures*.

The original cTAKES work demonstrated the benefits of combining clinical dictionaries, UMLS concept mapping and contextual attributes in a modular clinical NLP architecture [5]. *The current project supports the view that this approach can provide broad disease and terminology coverage when applied to complete discharge summaries*. cTAKES achieved higher recall and stronger overall ICD-10 coding performance, including higher coding-suggestion accuracy and correct-code coverage. However, its lower precision demonstrates the continuing risk that broad terminology matching will identify medically valid expressions that do not represent eligible patient morbidities.

The reported cTAKES results must be interpreted as the performance of the complete customised pipeline. Additional assertion safeguards, abbreviation recognition, exact UMLS terminology supplementation, semantic filtering and UK-specific mapping were applied around the standard cTAKES components. *The results should therefore not be interpreted as a benchmark of the unmodified cTAKES distribution*.

Previous research on Bio_ClinicalBERT and related biomedical transformers has shown that **domain-specific pretraining can support clinical entity extraction by providing representations adapted to biomedical and clinical language** [6], [12]. *The higher precision achieved by the transformer-centred pipeline is consistent with the ability of contextual representations to distinguish relevant disease mentions from surrounding text.*

Nevertheless, the *lower recall demonstrates that clinical pretraining alone does not ensure complete morbidity extraction*. Performance remained dependent on the size and coverage of the manually annotated fine-tuning corpus, the selected confidence threshold and the processing of long documents. Furthermore, the final Bio_ClinicalBERT-centred pipeline included medSpaCy ConText, section rules, semantic filtering and terminology linking. The observed precision should therefore be attributed to the complete transformer-centred pipeline rather than to Bio_ClinicalBERT alone.

MedCAT represents an existing example of combining statistical learning with terminology-based concept linking [9]. Its design supports the broader argument that contextual representations and medical terminologies can provide complementary information. *The present study contributes additional task-specific evidence for this argument by demonstrating different precision and recall profiles under the same morbidity definition and evaluation conditions*.

Research on automatic ICD coding frequently treats code prediction as a multi-label document-classification problem. CAML, for example, predicts a collection of ICD codes from a complete clinical document using label-specific attention [14]. BERT-based coding research has similarly considered the challenges of long documents, large label spaces and rare codes [13]. More recent work has also investigated generative models for discharge-summary coding [15]. *The present project used a different design*. It first extracted individual morbidity concepts, determined their clinical context and then produced a code suggestion associated with each retained morbidity. This provides a more explicit relationship between the clinical evidence and the suggested code. It also permits separate measurement of extraction performance, coding coverage and coding-suggestion accuracy.

The mapping results are consistent with NHS and SNOMED guidance that mapping between SNOMED CT and ICD-10 codes is not always a simple one-to-one operation [2], [16]. *Some mappings depended on information about disease type, stage, cause, anatomical site or complications*. The use of ranked candidates and review indicators was therefore more appropriate than presenting every mapping as definitive.

Direct numerical comparison with F1 scores or coding-suggestion accuracy reported by other studies would not be appropriate because the datasets, entity definitions, annotation policies, code systems and matching rules differ. *The main contribution of this project is the controlled head-to-head comparison of two substantially different end-to-end approaches on the same complete records, using a common concept-level gold standard and the same evaluation procedure.*

6.4 Limitations and future work

6.4.1 Limitations

A few limitations should be considered when interpreting the results.

1. Although the expanded evaluation included 200 previously unseen discharge summaries and 2,103 gold-standard morbidity concepts, all clinical records originated from MIMIC-IV and therefore represent documentation from a single United States healthcare environment. Documentation practices, abbreviations, terminology and coding conventions may differ across institutions and healthcare systems, particularly within the NHS. External validation is therefore required before the findings can be generalised beyond the present dataset.

2. The gold standard was based on manual review within this project rather than independent annotation by multiple clinical or coding experts. Although a consistent morbidity definition and review procedure were applied, independent annotation and inter-annotator agreement would provide stronger evidence for the reliability of the reference standard.

3. The two evaluated systems were customised end-to-end pipelines. Bio_ClinicalBERT was combined with contextual, semantic and terminology-based post-processing, while cTAKES was supplemented with abbreviation handling, assertion safeguards, terminology extensions and UK-specific mapping. The results therefore cannot be generalised directly to unmodified Bio_ClinicalBERT, EHRKit or cTAKES installations.

4. The Bio_ClinicalBERT model was developed using a relatively small manually annotated training corpus. Rare diseases, uncommon abbreviations and unusual clinical expressions may therefore have been insufficiently represented. In addition, processing long discharge summaries through overlapping windows may limit the model's ability to use information distributed across distant parts of a document.

5. The coding evaluation was conducted at the level of three-character NHS ICD-10 families. It therefore does not establish accuracy for complete four- or five-character coding, which may require additional information about disease subtype, anatomical site, severity, cause or complications.

6. Finally, both pipelines were evaluated as research prototypes rather than within a live clinical workflow. Runtime requirements, integration with operational EHR systems, user interaction, clinical-coder workload and the practical consequences of false-positive or false-negative outputs were outside the scope of this study..

6.4.2 Future work

- Future work should focus less on simply increasing the number of MIMIC-IV records and more on **external and independent validation**. Both pipelines should be evaluated on patient-disjoint records from multiple healthcare organisations, ideally including clinical text produced within the NHS. Such evaluation would help determine whether the precision–recall profiles observed in MIMIC-IV remain stable across different documentation styles, terminology and clinical settings.
- The gold standard should also be independently annotated by at least two reviewers with appropriate clinical or coding expertise, allowing inter-annotator agreement to be measured and disagreements to be resolved systematically.
- Bio_ClinicalBERT could be fine-tuned on a larger and more diverse annotated morbidity corpus. Increased representation of uncommon diseases, abbreviations and variable clinical expressions may improve recall while retaining the contextual selectivity demonstrated in the present evaluation.
- A further extension would be to introduce a dedicated coding model after concept normalisation. Rather than predicting unrestricted codes, such a model could rank only ICD-10 families supported by the relevant NHS terminology and mapping resources. Inputs could include the extracted morbidity, surrounding clinical context, linked

SNOMED CT concept and mapping attributes. The model should retain the ability to abstain or return a short ranked candidate list when the available information is insufficient for a reliable primary suggestion.

6.4.3 Hybrid morbidity-extraction pipeline

The complementary precision and recall profiles observed in the evaluation suggest that a hybrid pipeline is a promising direction. However, combining the outputs of the two systems cannot be assumed to improve performance automatically, because increased morbidity coverage could also introduce additional false positives. A hybrid architecture would therefore need to be implemented and evaluated independently on new held-out records.

One possible architecture would use cTAKES as a high-coverage candidate generator. Its disease mentions, UMLS CUIs, assertion attributes and terminology variants could be transferred to a Python-based processing layer using JSON, CSV or a service interface. A transformer component based on Bio_ClinicalBERT could then be adapted to examine each candidate together with its surrounding clinical context and determine whether it represents an eligible patient morbidity.

The transformer-based component could reject inappropriate terminology matches caused by negation, uncertainty, family history, ambiguous abbreviations or non-diagnostic sections. Independently predicted high-confidence Bio_ClinicalBERT spans could also be retained when they did not overlap a cTAKES candidate. The resulting concepts would then be merged, normalised and deduplicated before ICD-10 coding.

Figure 6.1 presents the proposed hybrid process.

The use of Python for Bio_ClinicalBERT and Java for cTAKES does not prevent such integration. The two components could exchange mention-level information through a language-independent representation containing text offsets, terminology concepts, confidence scores and contextual attributes.

The hybrid system should be evaluated against each individual pipeline using a new patient-disjoint gold standard. In addition to precision, recall and F1, the evaluation should measure how frequently one component recovers an error made by the other, whether contextual validation successfully removes false positives, and whether the additional filtering introduces new false negatives.

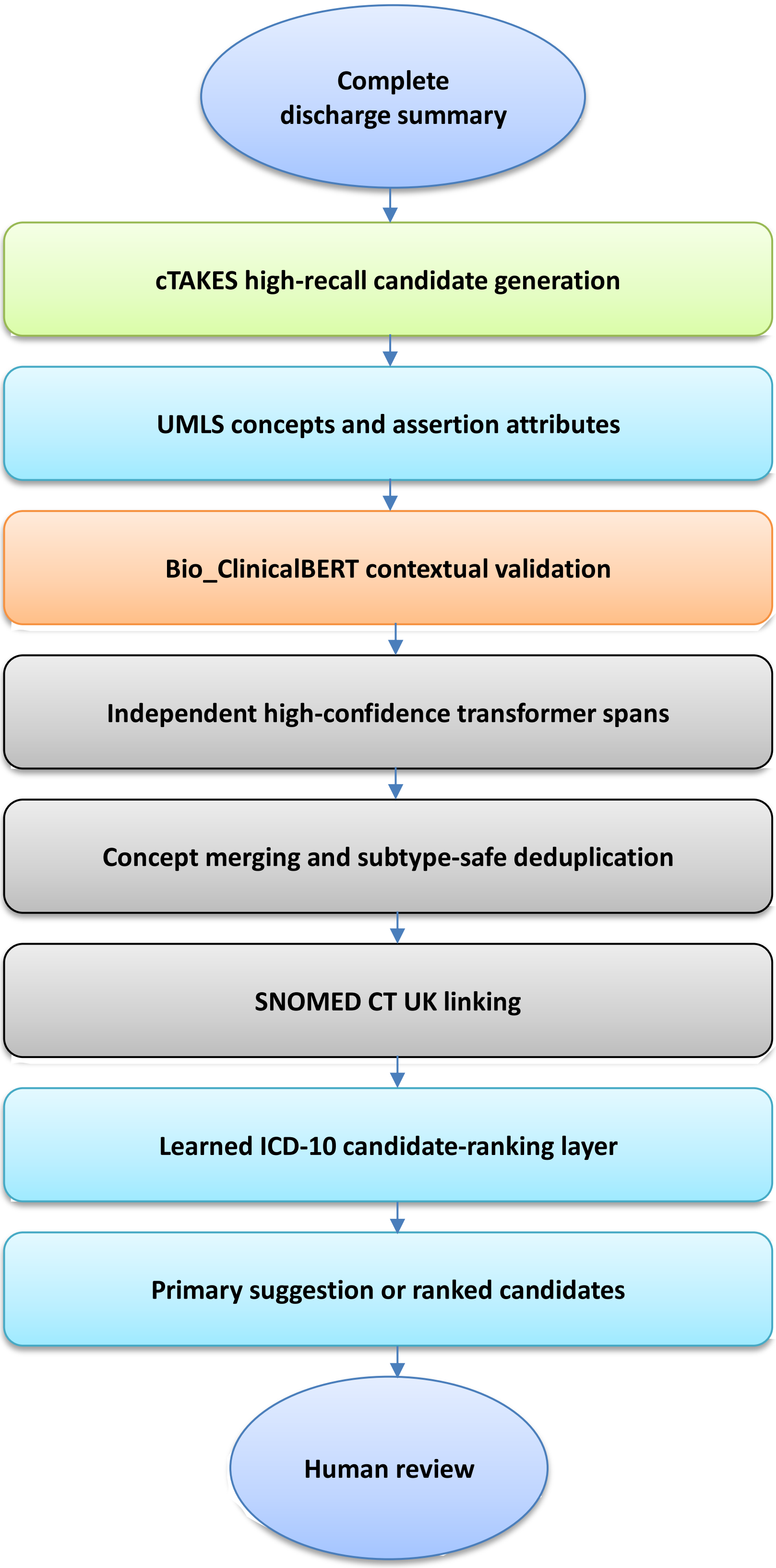

Figure 6.1 Proposed hybrid morbidity-extraction and coding architecture

6.5 Conclusion

This dissertation compared a transformer-centred Bio_ClinicalBERT pipeline with a terminology-centred cTAKES pipeline for extracting unique patient morbidities from complete MIMIC-IV discharge summaries and assigning suggested NHS ICD-10 5th Edition families.

The expanded evaluation confirmed distinct operating profiles for the two approaches. Bio_ClinicalBERT achieved substantially higher precision and a modestly higher F1 score, whereas cTAKES achieved higher recall and stronger ICD-10 coding performance. Rather than demonstrating uniform superiority of either approach, the results showed that the two pipelines offered different and complementary strengths: greater contextual selectivity from the transformer-centred workflow and broader morbidity and terminology coverage from the cTAKES-centred workflow.

The study demonstrates that free-text discharge summaries can be transformed into structured, concept-level morbidity information and associated with standardised coding suggestions. It also demonstrates the importance of evaluating morbidity extraction and ICD-10 assignment separately, since successful identification of a clinical condition does not necessarily guarantee that an appropriate coding suggestion can be produced.

Taken together, the quantitative results and practical experience with both toolkits support further investigation of a hybrid architecture combining terminology-based candidate generation with transformer-based contextual validation and subsequent ICD-10 candidate ranking. Such an approach could potentially exploit the complementary strengths of both pipelines, but its effectiveness would need to be established through independent implementation and external evaluation on newly annotated clinical records.

References

[1] Johnson, A., Pollard, T., Horng, S., Celi, L. A. and Mark, R., "MIMIC-IV-Note: Deidentified free-text clinical notes," PhysioNet, version 2.2, 2023. Link: https://doi.org/10.13026/1n74-ne17

[2] NHS Digital. *SCCI0034 SNOMED CT: Implementing the Standard*, version 1.0, final release, 2020. Link: https://digital.nhs.uk/binaries/content/assets/website-assets/isce/scci0034/0034352016guidance-v2.pdf

[3] NHS Digital. "NHS ICD-10 5th Edition data files." Terminology Reference-data Update Distribution. Accessed: 7 August 2026. Link: https://isd.digital.nhs.uk/trud/users/guest/filters/0/categories/28/items/258/licences

[4] Bodenreider, O. "The Unified Medical Language System (UMLS): integrating biomedical terminology." Nucleic Acids Research, vol. 32, suppl. 1, pp. D267–D270, 2004. Link: https://doi.org/10.1093/nar/gkh061

[5] Savova, G. K., Masanz, J. J., Ogren, P. V., Zheng, J., Sohn, S., Kipper-Schuler, K. C. and Chute, C. G. "Mayo Clinical Text Analysis and Knowledge Extraction System (cTAKES): architecture, component evaluation and applications." Journal of the American Medical Informatics Association, vol. 17, no. 5, pp. 507–513, 2010. Link: https://doi.org/10.1136/jamia.2009.001560

[6] Alsentzer, E., Murphy, J. R., Boag, W., Weng, W.-H., Jin, D., Naumann, T. and McDermott, M. B. A. "Publicly available clinical BERT embeddings." *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, Minneapolis, Minnesota, USA, pp. 72–78, 2019. Link: https://doi.org/10.18653/v1/W19-1909

[7] Li, I., You, K., Qiao, Y., Huang, L., Hsieh, C.-C., Rosand, B., Goldwasser, J. and Radev, D. "EHRKit: A Python Natural Language Processing Toolkit for Electronic Health Record Texts." *arXiv preprint arXiv:2204.06604*, 2022. Link: https://doi.org/10.48550/arXiv.2204.06604

[8] Lossio-Ventura, J. A., Sun, R., Boussard, S. and Hernandez-Boussard, T. "Clinical concept recognition: Evaluation of existing systems on EHRs." Frontiers in Artificial Intelligence, vol. 5, article 1051724, 2023. Link: https://doi.org/10.3389/frai.2022.1051724

[9] Kraljevic, Z., Searle, T., Shek, A., Roguski, L., Noor, K., Bean, D., et al. "Multi-domain clinical natural language processing with MedCAT: The Medical Concept Annotation Toolkit." Artificial Intelligence in Medicine, vol. 117, article 102083, 2021. Link: https://doi.org/10.1016/j.artmed.2021.102083

[10] Chapman, W. W., Bridewell, W., Hanbury, P., Cooper, G. F. and Buchanan, B. G. "A simple algorithm for identifying negated findings and diseases in discharge summaries." Journal of Biomedical Informatics, vol. 34, no. 5, pp. 301–310, 2001. Link: https://doi.org/10.1006/jbin.2001.1029

[11] Harkema, H., Dowling, J. N., Thornblade, T. and Chapman, W. W. "ConText: An algorithm for determining negation, experiencer, and temporal status from clinical reports." Journal of Biomedical Informatics, vol. 42, no. 5, pp. 839–851, 2009. Link: https://doi.org/10.1016/j.jbi.2009.05.002

[12] Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H. and Kang, J. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining." *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020. Link: https://doi.org/10.1093/bioinformatics/btz682

[13] Pascual, D., Luck, S. and Wattenhofer, R. "Towards BERT-based automatic ICD coding: limitations and opportunities." *Proceedings of the 20th Workshop on Biomedical Language Processing*, pp. 54–63, 2021. Link: https://doi.org/10.18653/v1/2021.bionlp-1.6

[14] Mullenbach, J., Wiegreffe, S., Duke, J., Sun, J. and Eisenstein, J. "Explainable prediction of medical codes from clinical text." *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, vol. 1, pp. 1101–1111, 2018. Link: https://doi.org/10.18653/v1/N18-1100

[15] Falis, M., Gema, A. P., Dong, H., Daines, L., Basetti, S., Holder, M., Penfold, R. S., Birch, A. and Alex, B. "Can GPT-3.5 generate and code discharge summaries?" *Journal of the American Medical Informatics Association*, vol. 31, no. 10, pp. 2284–2293, 2024. Link: https://doi.org/10.1093/jamia/ocae132

[16] SNOMED International. "Mapping reference set." *SNOMED CT Glossary*. Accessed: 5 August 2026. Link: https://docs.snomed.org/snomed-international-documents/snomed-ct-glossary/m/mapping-reference-set

[17] Eyre, H., Chapman, A. B., Peterson, K. S., Shi, J., Alba, P. R., Jones, M. M., Box, T. L., DuVall, S. L. and Patterson, O. V. "Launching into clinical space with medspaCy: a new clinical text processing toolkit in Python." AMIA Annual Symposium Proceedings, vol. 2021, pp. 438–447, 2022. Link: https://pubmed.ncbi.nlm.nih.gov/35308962/